All India Aakash Test Series for NEET-2028

TEST - 2 (Code-C)

Click here for Code-D Sol.

Test Date : 25/10/2026

ANSWERS

1. (3)	37. (2)	73. (3)	109. (4)	145. (1)
2. (1)	38. (1)	74. (2)	110. (1)	146. (3)
3. (1)	39. (3)	75. (4)	111. (2)	147. (4)
4. (2)	40. (2)	76. (2)	112. (4)	148. (2)
5. (3)	41. (1)	77. (1)	113. (4)	149. (2)
6. (3)	42. (3)	78. (4)	114. (1)	150. (1)
7. (4)	43. (2)	79. (2)	115. (2)	151. (2)
8. (2)	44. (1)	80. (2)	116. (2)	152. (4)
9. (2)	45. (1)	81. (1)	117. (2)	153. (4)
10. (1)	46. (1)	82. (2)	118. (3)	154. (4)
11. (1)	47. (4)	83. (2)	119. (3)	155. (1)
12. (4)	48. (3)	84. (2)	120. (3)	156. (1)
13. (4)	49. (3)	85. (3)	121. (4)	157. (2)
14. (1)	50. (4)	86. (1)	122. (1)	158. (1)
15. (2)	51. (1)	87. (3)	123. (3)	159. (3)
16. (2)	52. (4)	88. (2)	124. (2)	160. (2)
17. (4)	53. (1)	89. (2)	125. (2)	161. (4)
18. (1)	54. (3)	90. (2)	126. (4)	162. (4)
19. (4)	55. (4)	91. (3)	127. (4)	163. (3)
20. (3)	56. (2)	92. (1)	128. (4)	164. (4)
21. (3)	57. (2)	93. (4)	129. (4)	165. (3)
22. (2)	58. (2)	94. (1)	130. (3)	166. (4)
23. (4)	59. (4)	95. (2)	131. (3)	167. (2)
24. (4)	60. (2)	96. (1)	132. (1)	168. (2)
25. (1)	61. (4)	97. (1)	133. (1)	169. (3)
26. (4)	62. (3)	98. (2)	134. (3)	170. (2)
27. (3)	63. (3)	99. (1)	135. (3)	171. (1)
28. (4)	64. (3)	100. (2)	136. (4)	172. (3)
29. (2)	65. (1)	101. (2)	137. (3)	173. (2)
30. (2)	66. (1)	102. (4)	138. (4)	174. (3)
31. (3)	67. (1)	103. (1)	139. (1)	175. (3)
32. (3)	68. (3)	104. (1)	140. (2)	176. (2)
33. (1)	69. (1)	105. (2)	141. (4)	177. (2)
34. (1)	70. (2)	106. (3)	142. (3)	178. (2)
35. (3)	71. (4)	107. (4)	143. (4)	179. (2)
36. (3)	72. (3)	108. (2)	144. (3)	180. (4)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint: Use $h = \frac{u^2 \sin^2\theta}{2g}$ and time of flight $T = \frac{2u\sin\theta}{g}$

Sol.: $T^2 = \frac{4u^2\sin^2\theta}{g^2}$

$\therefore\ h = \frac{u^2\sin^2\theta}{2g} = \frac{gT^2}{8}$

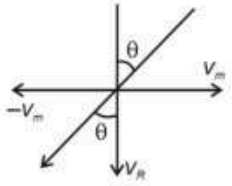
$= \frac{10\times 36}{8}$

= 45 m

2. Answer (1)

Hint: Use the concept of relative motion

Sol.:



$\overrightarrow{v_{Rm}} = \overrightarrow{v_R} - \overrightarrow{v_m}$

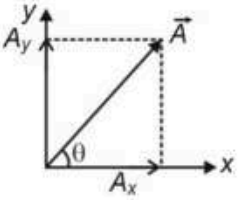
$\tan\theta = \frac{v_m}{v_R}$

$\tan\theta = \frac{8}{6} \Rightarrow \theta = 53°$

3. Answer (1)

Hint: Rectangular component of vector depends on angle θ made by vector with *x*-axis or *y*-axis.

Sol.:



$A_x = A\cos\theta$

$A_y = A\sin\theta$

On changing value of θ, A_x and A_y will change.

A vector can be split into infinite number of components.

4. Answer (2)

Hint: Use $\cos\theta = \frac{d_x}{|\vec{d}|}$

Sol.: $\vec{d} = (3-1)\hat{i} + (5-2)\hat{j} + (2-1)\hat{k}$

$|\vec{d}| = \sqrt{(2^2 + 3^2 + 1^2)}$

$= \sqrt{14}$ units

$d_x = 2$ units

$\cos\theta = \left(\frac{2}{\sqrt{14}}\right)$

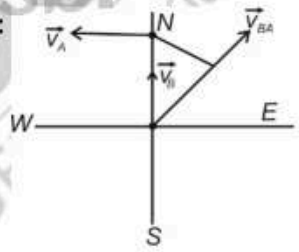
$\Rightarrow\ \theta = \cos^{-1}\left(\frac{2}{\sqrt{14}}\right)$

5. Answer (3)

Hint: Use concept of relative motion.

$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A}$

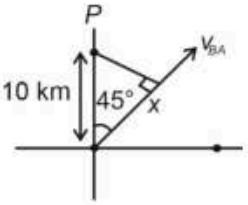
Sol.:



$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A}$

$\overrightarrow{v_{BA}} = (20\hat{i} + 20\hat{j})$ km/h

$|\overrightarrow{v_{BA}}| = 20\sqrt{2}$ km/h



$x = 5\sqrt{2}$ km

$t = \frac{x}{|v_{BA}|} = \frac{5\sqrt{2}}{20\sqrt{2}}$

$= \frac{1}{4}$h = 15 min

6. Answer (3)

Hint: Use $S = ut + \frac{1}{2}at^2$ in vector form

Sol.: $\vec{x} - \vec{x}_0 = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$\vec{x} - \vec{x}_0 = \frac{1}{2}\left(2\hat{i} + 3\hat{k}\right)(2)^2$

$\vec{x} - \vec{x}_0 = 4\hat{i} + 6\hat{k}$

$\vec{x} - (\hat{j} + 3\hat{k}) = 4\hat{i} + 6\hat{k}$

$\vec{x} = (4\hat{i} + \hat{j} + 9\hat{k})$ m

7. Answer (4)

Hint: Horizontal component of velocity will remain same.

Sol.: $v_y = u_y - gt$

$v_y = 40 - 10t$, $v_x = 10$ m/s

$\tan 45^\circ = \left(\frac{v_y}{v_x}\right)$

$\frac{40 - 10t}{10} = 1$

$10t = 40 - 10$

$t = 3$ s

8. Answer (2)

Hint: Use range $R = \frac{u^2 \sin 2\theta}{g}$

Sol.: $\frac{\left(\frac{v}{2}\right)^2 \sin 90^\circ}{g} = \frac{v^2 \sin 2\theta}{g}$

$\theta = \frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$

9. Answer (2)

Hint: Compare equation of trajectory $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$ with given equation.

Sol.: On comparing,

$\tan\theta = \sqrt{3} \Rightarrow \theta = 60^\circ$

$u^2\cos^2\theta = 1 \Rightarrow u = 2$ m/s

$T = \frac{2u\sin\theta}{g}$

$= \frac{2 \times 2\sin 60^\circ}{10}$

$= \frac{\sqrt{3}}{5}$ s

10. Answer (1)

Hint & Sol.: Magnitude of horizontal component and vertical component of velocity, are same at both point of projection and landing point. Therefore, speed at both the points will be same.

11. Answer (1)

Hint: Resultant $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Sol.: For parallel vectors, $\theta = 0^\circ$

$\Rightarrow R = A + B$

For anti-parallel vectors, $\theta = 180^\circ$

$\Rightarrow R = A - B$

For $\theta = 60^\circ$,

$R = \sqrt{A^2 + B^2 + AB}$

For $\theta = 120^\circ$,

$R = \sqrt{A^2 + B^2 - AB}$

$\therefore R_a > R_c > R_d > R_b$

12. Answer (4)

Hint: Acceleration of object w.r.t, observer $\vec{a}_{PO} = -2g\hat{j}$, with $\hat{j}$ along vertically upward direction.

Sol.: $x = v_0 t$

$y = -\frac{1}{2}2gt^2$

$y = \frac{-gx^2}{v_0^2}$ is equation of parabola.

13. Answer (4)

Hint & Sol.:

In any triangle, length of the third side is always less than the sum of the lengths of other two sides but greater than their difference.

14. Answer (1)

Hint: The relation between x and y gives the equation of trajectory.

Sol.: $\vec{a} = \left(2\hat{i} + 3\hat{j}\right)$ m/s^2

Writing $s = ut + \frac{1}{2}at^2$ along x-axis

$x = \frac{1}{2}(2)t^2$

$x = t^2$ ...(1)

Writing $s = \left(ut + \frac{1}{2}at^2\right)$ along y-axis

$y = \frac{3}{2}t^2$ ...(2)

$y = \frac{3}{2}x$

$2y = 3x$

15. Answer (2)

Hint: Time taken in quicker path, $t = \frac{d}{V_{BR}}$

Sol.: Time taken in shortest path,

$T = \frac{d}{\sqrt{V_{BR}^2 - V_R^2}}$

$5 + t = T$

$5 + \frac{d}{V_{BR}} = \frac{d}{\sqrt{V_{BR}^2 - V_R^2}}$

$\frac{d}{6} - \frac{d}{10} = 5$

$\frac{5d - 3d}{30} = 5$

$d = 75$ m

16. Answer (2)

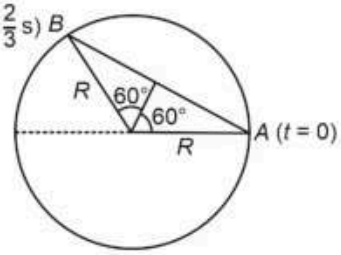
Hint & Sol.: In uniform circular motion, direction of velocity and direction of centripetal acceleration will both continuously change, but magnitude and direction of angular velocity will both remain same.

17. Answer (4)

Hint: Angular displacement = $\omega \times t$

$= \frac{2\pi}{3}$ rad = 120°

Sol.:



Displacement = $2R\cos 30°$

$= 2R \times \frac{\sqrt{3}}{2}$

$= \sqrt{3}R$

18. Answer (1)

Hint & Sol.: In oblique projectile motion, time of upward journey is equal to time taken for downward journey. At maximum height, speed will be minimum.

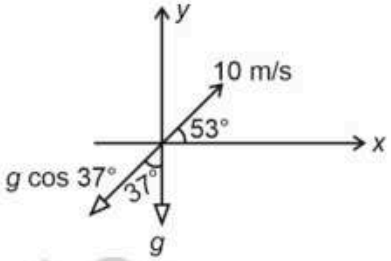
19. Answer (4)

Hint & Sol.: Displacement, velocity and acceleration all of them are vector quantities.

20. Answer (3)

Hint: The rate of change of speed is equal to tangential acceleration of object.

Sol.:



$a_T = g\cos 37°$

$= 8$ m/s^2

21. Answer (3)

Hint: Range $R = (u\cos\theta)\,T$

Sol.: $H_1 = H_2$

$u_1\sin\theta_1 = u_2\sin\theta_2 \Rightarrow T_1 = T_2$

$\sin\theta_1 > \sin\theta_2$ $(\because \theta_1 > \theta_2)$

$\Rightarrow$ $u_1 < u_2$ and $\cos\theta_1 < \cos\theta_2$

$R_1 = (u_1\cos\theta_1)\,T_1$

$R_2 = (u_2\cos\theta_2)\,T_2$

$\therefore R_1 < R_2$

22. Answer (2)

Hint: Displacement $d = \sqrt{H^2 + \left(\frac{R}{2}\right)^2}$

Sol.: $d = \sqrt{\left(\frac{u^2\sin^2\theta}{2g}\right)^2 + \left(\frac{u^2\sin 2\theta}{2g}\right)^2}$

$d = \frac{u^2}{2g}\sqrt{\sin^4\theta + \sin^2 2\theta}$

$d = \frac{u^2}{2g}\sqrt{\frac{1}{16} + \frac{3}{4}}$

$= \frac{u^2}{8g}\sqrt{13}$

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23. Answer (4)

Hint: Radius of curvature at maximum height of projectile,

$$R = \frac{(u\cos\theta)^2}{g}$$

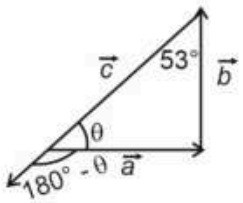
Sol.: $\frac{R}{H} = \frac{u^2\cos^2\theta}{g \times \frac{u^2\sin^2\theta}{2g}}$

$= 2\cot^2\theta = 2\times\left(\frac{1}{\sqrt{3}}\right)^2$

$= \frac{2}{3}$

24. Answer (4)

Hint & Sol.:



In this case, $\theta = 37°$

Required angle = $180° - \theta = 180 - 37°$

$= 143°$

25. Answer (1)

Hint & Sol.: When angle between velocity and acceleration is obtuse, then one component of acceleration (tangential acceleration) decreases the magnitude of velocity while other component of acceleration (normal acceleration) changes the direction of velocity.

26. Answer (4)

Hint & Sol.: Direction of angular velocity of object moving in $x - y$ plane on a circular path, may be along positive or negative z-direction depending upon sense of rotation.

27. Answer (3)

Hint: Use resultant $R = \sqrt{a^2 + b^2 + 2ab\cos\theta}$

Sol.: $\sqrt{10}p = \sqrt{(2p)^2 + \left(\sqrt{2}p\right)^2 + 2\times 2p\left(\sqrt{2}p\right)\cos\theta}$

$10p^2 = 6p^2 + 4\sqrt{2}p^2\cos\theta$

$4p^2 = 4\sqrt{2}p^2\cos\theta$

$\cos\theta = \frac{1}{\sqrt{2}}$

$\theta = 45°$

28. Answer (4)

Hint: Unit vector $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

Sol.: Required vector $\vec{b} = b\hat{A}$

$\hat{A} = \frac{2\hat{i} + 2\hat{j}}{\sqrt{2^2 + 2^2}}$

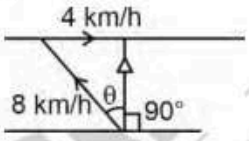
$\hat{A} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

$\vec{b} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)4\sqrt{2} = \left(4\hat{i} + 4\hat{j}\right)$

29. Answer (2)

Hint: To travel with minimum drift (along shortest path), angle made by boat with river flow should be obtuse.

Sol.:



$\sin\theta = \frac{4}{8} \Rightarrow \theta = 30°$

∴ Required angle = 90° + 30°

= 120°

30. Answer (2)

Hint: Time of flight $T = \frac{2u_y}{g}$, Range $R = u_x T$

Sol.: $\tan\theta = \frac{u_y}{u_x}$

$\tan\theta = \frac{4}{3}$

$\theta = 53°$

From vertical, $90° - \theta = 37°$

$T = \frac{2\times 40}{10} = 8$ s

$R = 30 \times 8 = 240$ m

$H = \frac{u_y^2}{2g} = \frac{(40)^2}{20} = 80$ m

31. Answer (3)

Hint & Sol.: In oblique projectile motion of an object, acceleration remains constant but speed continuously changes.

32. Answer (3)

Hint: Horizontal component of the velocity of object will remain same.

Sol.: $40 \cos 53° = v \cos 37°$

$40 \times \frac{3}{5} = v \times \frac{4}{5}$

$v = 30$ m/s

$v_y = v \sin 37°$

$= 30 \times \frac{3}{5}$

$= 18$ m/s

33. Answer (1)

Hint: y-component of velocity is given by $v_y = \frac{dy}{dt}$

Sol.: $y = 2x^2$

$y = 2t^4$

$\frac{dy}{dt} = 8t^3$

$v_y = 8 \times \left(\frac{1}{2}\right)^3$

$= 1$ m/s

34. Answer (1)

Hint: Resultant $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Sol.: $R = |\vec{C}|$

$|\vec{C}| = \sqrt{A^2 + B^2 + 2AB\cos 180°}$

$|\vec{C}| = \sqrt{A^2 + B^2 - 2AB}$

$= A - B$

$= |\vec{A}| - |\vec{B}|$

35. Answer (3)

Hint & Sol.: x-component, $R_x = R \cos\theta$

while y – component, $R_y = R \sin\theta$

$R_x \times R_y = R^2 \sin\theta \times \cos\theta$

$= \frac{R^2 \sin 2\theta}{2}$

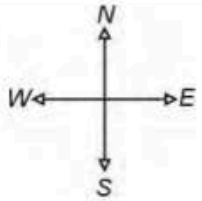
36. Answer (3)

Hint & Sol.: Magnitude of velocity and magnitude of acceleration are both constant in uniform circular motion, so change in magnitude will be zero.

37. Answer (2)

Hint: On multiplying a vector by (–1), the direction of the vector becomes opposite.

Sol.:



$\vec{d} = 5\hat{j}$ km

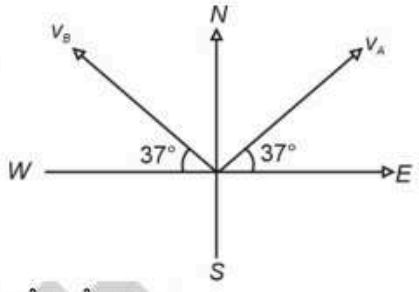
$\vec{d}_1 = -2 \times 5\hat{j}$

$= -10\hat{j}$ km (along south)

38. Answer (1)

Hint: Use relative velocity $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$

Sol.:



$\vec{v}_A = (8\hat{i} + 6\hat{j})$ m/s

$\vec{v}_B = (-8\hat{i} + 6\hat{j})$ m/s

$\vec{v}_{AB} = 8\hat{i} + 6\hat{j} + 8\hat{i} - 6\hat{j}$

$= 16\hat{i}$ m/s

39. Answer (3)

Hint: Net acceleration $a = \sqrt{a_x^2 + a_y^2}$

Sol.: $s = ut + \frac{1}{2}at^2$, along x direction gives

$6 = \frac{1}{2}a_x(2)^2 \Rightarrow a_x = 3$ m/s^2

$s = ut + \frac{1}{2}at^2$, along y-direction gives

$8 = \frac{1}{2}a_y(2)^2 \Rightarrow a_y = 4$ m/s^2

$a = \sqrt{3^2 + 4^2}$

$= 5$ m/s^2

40. Answer (2)

Hint: For $\vec{a} \parallel \vec{b}$, we have $\frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z}$

Sol.: $\vec{a} = 2\hat{i} + 3\hat{j} - 2\hat{k}$

$\vec{b} = \hat{i} + \lambda\hat{j} + \mu\hat{k}$

$\therefore 2 = \frac{3}{\lambda} = \frac{-2}{\mu}$

$\Rightarrow \frac{\lambda}{\mu} = -\frac{3}{2}$

41. Answer (1)

Hint: For collision, vertical component of velocity of both particles should be same.

Sol.: $v_A \sin\theta = v_B$

$20 \sin\theta = 10$

$\sin\theta = \frac{1}{2}$

$\theta = 30°$

$x = v_A \cos 30° \times t$

$= 20 \times \frac{\sqrt{3}}{2} \times \frac{1}{2}$

$= 5\sqrt{3}$ m

42. Answer (3)

Hint: Compare given equation $y = \frac{x}{\sqrt{3}} - \frac{gx^2}{6}$ with

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

Sol.: $\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = 30°$

$2u^2\cos^2\theta = 6$

$u^2 \times \frac{3}{4} = 3$

$u = 2$ m/s

43. Answer (2)

Hint: Velocity, $\vec{v} = \frac{d\vec{r}}{dt}$

Sol.: $\vec{v} = 3\hat{i} + 6\hat{j}$

$\Rightarrow$ Velocity along y-axis = 6 unit.

44. Answer (1)

Hint: Split $\overrightarrow{OB}$ in x-y rectangular components.

Sol.: $\vec{R} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$

$= r\hat{i} + r\cos 45°\hat{i} + r\sin 45°\hat{j} + r\hat{j}$

$= \left(r + \frac{r}{\sqrt{2}}\right)\hat{i} + \left(r + \frac{r}{\sqrt{2}}\right)\hat{j}$

$= (\sqrt{2}r + r)(\cos 45°\hat{i} + \sin 45°\hat{j})$

$\vec{R} = \left(\sqrt{2}r + r\right)$ along $\overrightarrow{OB}$

45. Answer (1)

Hint & Sol.: $\sin(\pi t) = \frac{x}{a}$ ...(i)

$\cos(\pi t) = \frac{y}{b}$ ...(ii)

On squaring (i) and (ii), and then adding,

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \sin^2(\pi t) + \cos^2(\pi t)$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2} = 1 \quad (\because b = a)$

$\Rightarrow x^2 + y^2 = a^2$, is equation of circle.

[CHEMISTRY]

46. Answer (1)

Hint: For alkali metals, down the group, negative electron gain enthalpy decreases.

Sol.:

Element	$\Delta_{eg}H$ (kJ mol^{-1})
Na	– 53
K	– 48
S	– 200
Se	– 195

47. Answer (4)

Hint: Samarium is lanthanoid metal

Sol.:

Vanadium	–	Transition metal
Samarium	–	Lanthanoid
Chlorine	–	Non-metal
Germanium	–	Metalloid

48. Answer (3)

Hint: IUPAC name of the element having atomic number 101 is Unnilunium.

Sol.:

Atomic number	Name according to IUPAC nomenclature
101	Unnilunium
109	Unnilennium

49. Answer (3)

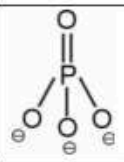
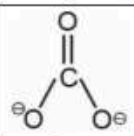
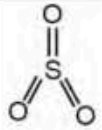
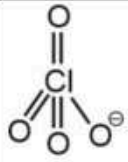
Hint: Electronic configuration of element with atomic number 49 is $[Kr]4d^{10}5s^25p^1$

Sol.: The element is Indium and it belongs to 5th period and 13th group.

50. Answer (4)

Hint: The elements which contain d-orbitals can form $p\pi = d\pi$ bond

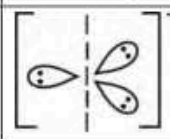
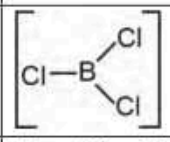
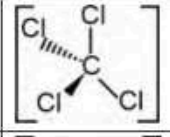
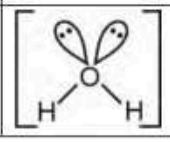
Sol.:

•		One pπ – dπ bond between P and O
•		It contains only pπ – pπ bond
•		It contains one pπ – pπ bond and two pπ – dπ bonds.
•		It contains one pπ – dπ bond.

51. Answer (1)

Hint: In I_3^-, central I atom is sp^3d hybridised.

Sol.:

Species	Arrangement of electron pairs	Bond angle
I_3^-		180°
BCl_3		120°
CCl_4		109.5°
H_2O		104.5°

52. Answer (4)

Hint: Shape of XeF_4 is square planar

Sol.:

Molecule	Shape	Hybridization of central atom
SF_4	See-saw	sp^3d
XeF_4	Square planar	sp^3d^2
XeO_3	Pyramidal	sp^3
SO_2	Bent	sp^2

53. Answer (1)

Hint: If bond order is zero then the species does not exist

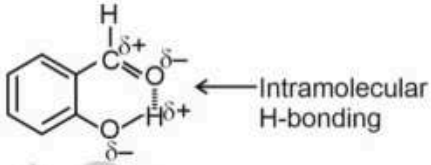
bond order = $\frac{1}{2}(N_b - N_a)$.

Sol.: • Bond order of He_2 and Be_2 are zero.

• B_2 : $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2\left(\pi 2p_x^1 = \pi 2p_y^1\right)$. It is paramagnetic in nature due to the presence of 2 unpaired electrons.

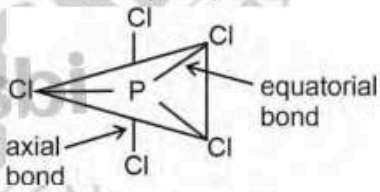
54. Answer (3)

Hint: Intramolecular hydrogen bond is formed when hydrogen atom is in between the two highly electronegative (F, O, N) atoms present within the same molecule.

Sol.: 

55. Answer (4)

Hint: P in PCl_5 in sp^3d hybridised

Sol.: 

- Axial bond pairs suffer more repulsive interaction from the equatorial bond pairs, therefore axial bonds have been found to be slightly longer and hence slightly weaker than the equational bonds.
- Maximum four atoms lie in a plane

56. Answer (2)

Hint: Bond order = $\frac{1}{2}(N_b - N_a)$

Sol.:

Species	Bond order
O_2	$\frac{1}{2}(10-6)=2$
O_2^-	$\frac{1}{2}(10-7)=1.5$
O_2^{2-}	$\frac{1}{2}(10-8)=1$
O_2^+	$\frac{1}{2}(10-5)=2.5$

57. Answer (2)

Hint: If net-dipole moment is non-zero then the molecule will be polar

Sol.:

Molecule	Shape	Dipole moment (μ)
PCl_3	Pyramidal (Non-planar)	$\mu \neq 0$
H_2O	Bent (planar)	$\mu \neq 0$
$CHCl_3$	Tetrahedral (Non-planar)	$\mu \neq 0$
NF_3	Pyramidal (Non-planar)	$\mu \neq 0$
BF_3	Trigonal planar	$\mu = 0$

58. Answer (2)

Hint: NO_2 is an odd electron molecule

Sol.: • Octet rule is not satisfied for all the atoms of NO_2 $\ddot{O}=\overset{\oplus}{\ddot{N}}-\ddot{O}^{\ominus}$

• In SCl_2, Octet rule is obeyed by each atom $:\ddot{Cl}-\ddot{S}-\ddot{Cl}:$

59. Answer (4)

Hint: Smaller the size of atoms, smaller is the bond length.

Sol.:

Bond type	Covalent bond length (pm)
O — H	96
C — H	107
C = N	138
C = O	121

60. Answer (2)

Hint: Percentage ionic character = $\frac{\mu_{observed}}{\mu_{calculated}} \times 100$

Sol.: 1D = 3.33×10^{-30} Cm

Percentage ionic character

$$= \frac{1.5 \times 3.33 \times 10^{-30} \times 100}{120 \times 10^{-12} \times 1.6 \times 10^{-19}} = 26\ \%$$

61. Answer (4)

Hint: Electronic configuration of N_2 :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^2$

Sol.: Electronic configuration of N_2^+ :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^1$

Number of bonding electrons = 9

• Electronic configuration of F_2 :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2$

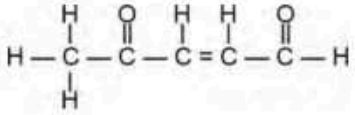
$(\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^2 = \pi^* 2p_y^2)$

Number of antibonding electrons = 8

62. Answer (3)

Hint: –CHO contains one π bond

Sol.:



Number of σ bonds = 12

Number of π bonds = 3

63. Answer (3)

Hint: Dipole moment depends on bond moments, orbital dipole and shape of the molecules.

Sol.:

Molecules	Dipole moment (D)
NH_3	1.47
NF_3	0.23
$CHCl_3$	1.04
CCl_4	0

64. Answer (3)

Hint: According to modern periodic law, the properties of elements are periodic function of their atomic numbers.

Sol.: Neptunium having atomic number 93 belongs to actinoid series.

65. Answer (1)

Hint: In OF_2 oxidation state of F is –1

Sol.: In OF_2 oxidation state of O will be + 2 whereas in Na_2O, the oxidation state of Na is +1 hence oxidation state of O will be –2.

66. Answer (1)

Hint: IUPAC official name of element having atomic number 105 is Dubnium

Sol.:

Atomic Number	IUPAC official name
108	Hassium
102	Nobelium
104	Rutherfordium

67. Answer (1)

Hint: Along the period, atomic size decreases

Sol.: • Down the group, atomic size generally increases.

- Correct order of atomic size

  Si > P > S > C > N

68. Answer (3)

Hint: If hybridisation of the central atom and shape of the species are same then they are called isostructural species.

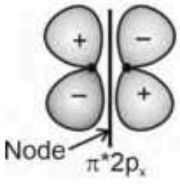
Sol.:

Species	Total no of electron	Hybridisation of central atom	Shape
NO_3^-	32	sp^2	Trigonal planar
CO_3^{2-}	32	sp^2	Trigonal planar
XeF_2	72	sp^3d	Linear
IF_2^-	72	sp^3d	Linear
ClO_4^-	50	sp^3	Tetrahedral
PO_4^{3-}	50	sp^3	Tetrahedral
NF_3	34	sp^3	Pyramidal
ClF_3	44	sp^3d	T-shape

69. Answer (1)

Hint: Molecular orbitals obtained from $2p_x$ and $2p_y$ orbitals are not symmetrical around the bond axis.

Sol.:

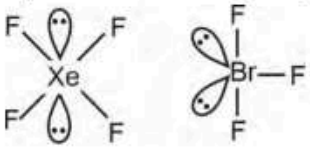


π^* antibonding MO has a node between the nuclei.

70. Answer (2)

Hint: Hybridization of Br in BrF_3 is sp^3d.

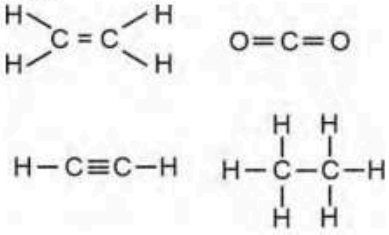
Sol.:



71. Answer (4)

Hint: If the hybridisation of carbon is sp^3, the hydrocarbon will not form a π-bond.

Sol.:



In C_2H_6, only σ – bonds are present.

72. Answer (3)

Hint: hybridization of I in IF_7 is sp^3d^3.

Sol.:

Molecule	Hybridisation of central atom	Shape
IF_5	sp^3d^2	Square pyramidal
IF_7	sp^3d^3	Pentagonal bipyramidal

73. Answer (3)

Hint: Electronic configuration of B_2 :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^1 = \pi 2p_y^1)$

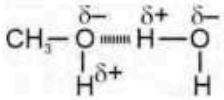
Sol.: Electronic configuration of B_2^- :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^1)$

74. Answer (2)

Hint: Methyl alcohol (CH_3OH) can form hydrogen bond with each other and also with water.

Sol.: Solubility of CH_3OH in water is explained by intermolecular hydrogen bonding.



75. Answer (4)

Hint: N_2O is a neutral oxide.

Sol.:

Compound	Chemical Nature
As_2O_3	Amphoteric
SO_2	Acidic
CaO	Basic

76. Answer (2)

Hint: Along the period, electronegativity generally increases.

Sol.:

Element	Electronegativity values (on Pauling scale)
Be	1.5
B	2.0
C	2.5
Mg	1.2

77. Answer (1)

Hint: f-block elements are called inner transition elements.

Sol.: • Bismuth is a metal while tellurium is a semi-metal.

- *s* and *p* block elements are main group elements.
- Vanadium is a *d*-block element.

78. Answer (4)

Hint: In Doberiener's triads, the middle element of each of the triads has an atomic weight about half way between the atomic weights of the other two.

Sol.: The Dobereiner's Triads are

- Li, Na, K
- Ca, Sr, Ba
- Cl, Br, I

79. Answer (2)

Hint: In general ionisation enthalpy increases along the period.

Sol.: • For stable electronic configuration the ionisation enthalpy of the element is higher than the element next to it in a period.

• For stable electronic configuration the ionisation enthalpy of N is higher than O.

80. Answer (2)

Hint: It is not necessary that only half filled orbitals participate in hybridisation.

Sol.: Promotion of electron is not an essential condition prior to hybridisation.

81. Answer (1)

Hint: For same bond order, large is the size of atom, larger is the bond length.

Sol.: Bond order of NO $= \frac{1}{2}(N_b - N_a)$

$= \frac{1}{2}(10-5) = 2.5$

Molecule	Bond energy (kJ mol^{-1})
B_2	293
F_2	155

82. Answer (2)

Hint: For triatomic species if the central atom is *sp* or sp^3d hybridised, the species will be linear in shape.

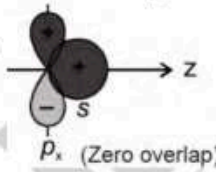
Sol.:

Species	Hybridisation of central atom	Shape
$[I_3]^+$	sp^3	Bent
SO_2	sp^2	Bent
$BeCl_2$	*sp*	Linear
NO_2^+	*sp*	Linear
H_2O	sp^3	Bent
$[ICl_2]^-$	sp^3d	Linear
CO_2	*sp*	Linear

83. Answer (2)

Hint: In zero overlap, net overlap of orbital is zero.

Sol.: •

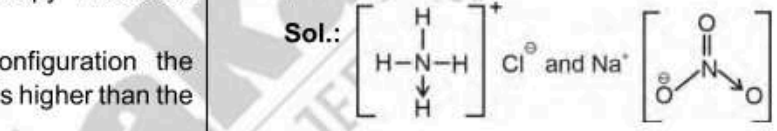


• (a) and (b) are negative overlap.

• (c) is the presentation of positive overlap.

84. Answer (2)

Hint: Coordinate bond is formed by sharing a lone pair to the vacant orbital of another atom.

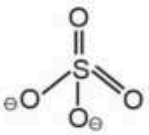
Sol.:  and Na^+

Contain both co-ordinate bond and ionic bonds.

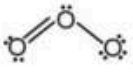
85. Answer (3)

Hint: Formal charge

$$= \left[\begin{array}{l}\text{Total number}\\ \text{of valence}\\ \text{electrons in}\\ \text{the free atom}\end{array}\right] - \left[\begin{array}{l}\text{Total number}\\ \text{of non bonding}\\ \text{electrons}\end{array}\right] - \frac{1}{2}\left[\begin{array}{l}\text{Total number}\\ \text{of bonding}\\ \text{electrons}\end{array}\right]$$

Sol.: Formal charge on S in 

$= \left[6 - 0 - \frac{1}{2}\times 12\right] = 0$

Formal charge on central O in 

$$= \left[6 - 2 - \frac{1}{2} \times 6\right] = 1$$

86. Answer (1)

Hint: For a given anion, smaller is the size of cation and greater the charge, greater will be the covalent character of ionic bond.

Sol.: • Correct order of ionic size :

$$Na^+ > Mg^{2+} > Al^{3+}$$

• Correct order of covalent character

$AlCl_3 > MgCl_2 > NaCl$

87. Answer (3)

Hint: Isoelectronic species contain same number of electrons.

Sol.: Cl^-, Ar, K^+ and Sc^{3+} contain 18 electrons each, hence these are isoelectronic species.

88. Answer (2)

Hint: If a system absorb energy, then the process is called endothermic.

Sol.: To ionise an atom, energy is always required to remove an electron from the atom.

89. Answer (2)

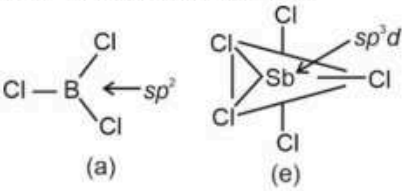
Hint: Element (X) is aluminium

Sol.: X_2Y_3 type of compound will be formed if Y belongs to oxygen family. Hence Y is oxygen having electronic configuration $1s^2 2s^2 2p^4$.

90. Answer (2)

Hint: 120° bond angle is observed in species having sp^2 or sp^3d hybridisation

Sol.:



Both (a) and (e) contain 120° bond angles.

[BOTANY]

91. Answer (3)

Hint: c-Prophase

e-Metaphase.

Sol.: The correct sequence of events is

$c \rightarrow e \rightarrow a \rightarrow b \rightarrow d$

92. Answer (1)

Hint: A – Family, B – Order, C – Class, D – Division/phylum

Sol.: The similar characters found in the members of an order are less in number as compared to different genera included in a family. Order primata is placed in class mammalia along with order carnivora.

93. Answer (4)

Hint: Mitosis restores the nucleo-cytoplasmic ratio.

Sol.: G_1 phase – Synthesis of proteins, RNA and ATP, except DNA resulting in cell growth.

S phase – Replication of DNA (polymer of deoxyribonucleotides)

G_2 phase – Synthesis of tubulin.

M phase – Helps in restoration of nucleo-cytoplasmic ratio after cell growth.

94. Answer (1)

Hint: As we go higher from species to kingdom, the number of common characteristics goes on decreasing.

Sol.: • Diptera (order) > Insecta (Class) (in terms of number of common characters)

• Chordata (Phylum) > Mammalia (Class) (in terms of number of members).

• Muscidae (Family) < Dicotyledonae (Class) (in terms of rank in taxonomical hierarchy)

• Anacardiaceae (Family) > Sapindales (Order) (in terms of number of common characters)

95. Answer (2)

Hint: Number of centromeres found in telophase are equal to the number of chromosomes found in a daughter cell obtained after cytokinesis.

Sol.: Number of chromosomes in a daughter cell

= Number of centromere

= 24

96. Answer (1)

Hint: Terminalisation of chiasmata can be observed during the final stage of prophase I.

Sol.: Terminalisation of chiasmata can be observed during diakinesis.

97. Answer (1)

Hint: Rudolf Virchow proposed *'Omnis cellula-e-cellula'*.

Sol.: During mitosis, the main function of the spindle fibres is to separate the chromosomes.

98. Answer (2)

Hint: DNA replication takes place in S-phase.

Sol.: G_1 phase occurs before S-phase, hence duplicated DNA is not seen here.

99. Answer (1)

Hint: Cell cycle in human completes in a day.

Sol.: Human cell in culture can progress through the cell cycle in only about 24 hours.

100. Answer (2)

Hint: The correct taxonomical hierarchy is as follows:

Species → Genus → Family → Order → Class → Division → Kingdom.

Sol.: Taxonomical categories in ascending order are as follows:

Triticum→	Poaceae→	Poales→	Monocotyledonae→	Angiospermae
(Genus)	(Family)	(Order)	(Class)	(Division)

101. Answer (2)

Hint: The number of pollen grains formed = n/4.

Sol.: The number of pollen grains formed = 100/4 = 25.

102. Answer (4)

Hint: Meiosis II occurs in haploid cells.

Sol.: On completion of meiosis II, the produced daughter cells have a different genetic composition than the parents.

103. Answer (1)

Hint: A mitotic metaphase chromosome is a typical X-shaped structure.

Sol.: A mitotic metaphase chromosome are made up of two sister chromatids held together by a centromere.

104. Answer (1)

Hint: Generally, order and other higher taxonomic categories are identified based on the aggregates of characters.

Sol.: Gibbons, based on the common features like presence of notochord and dorsal hollow neural system, are included in phylum chordata.

105. Answer (2)

Hint: Ernst Mayr has been called 'The Darwin of the 20th century'.

Sol.: Ernst Mayr was awarded with the triple crown of biology.

106. Answer (3)

Hint: Cell division is a process by which cell divides in two.

Sol.: The sequence of events by which a cell duplicates its genome, synthesise the other constituents of the cell and eventually divided into two daughter cells is termed cell cycle.

107. Answer (4)

Hint: Splitting of centromere ensures proper segregation of chromatids to each daughter cell.

Sol.: In animal cells, cytokinesis occurs in a centripetal manner.

108. Answer (2)

Hint: It marks the beginning of the second phase of mitosis.

Sol.: The complete disintegration of the nuclear envelope marks the start of metaphase.

109. Answer (4)

Hint: Carolus Linnaeus proposed binomial nomenclature.

Sol.: Ernst Mayr gave the definition of biological concept of species.

110. Answer (1)

Hint: Taxa can indicate categories of very different levels.

Sol.: Dog is a mammal and mammals are animals. Therefore, 'animals', 'mammals', and 'dogs' represent taxa at different levels.

111. Answer (2)

Hint: Lower the taxa, more are the characteristics that the members within the taxon share.

Sol.: Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

112. Answer (4)

Hint: In all organisms, the cell cycle is divided into two basic phases.

Sol.: Onion root tip cell has both interphase and M phase. In some lower plants and in some social insects haploid cells also divide by mitosis.

113. Answer (4)

Hint: G_0 phase of the cell cycle is metabolically active.

Sol.: Cells enter the G_1 phase post the G_0 phase.

114. Answer (1)

Hint: Centripetal mode of cytokinesis can be seen in animal cells.

Sol.: In animal cell, the cytokinesis is facilitated by furrow formation.

115. Answer (2)

Hint: Taxonomy includes classification identification and nomenclature.

Sol.: The taxonomic categories are distinct biological entities and not merely morphological aggregates.

116. Answer (2)

Hint: Both cat and gorilla belong to class mammalia.

Sol.: Both cat and gorilla share three taxonomical categories, namely class (Mammalia), Phylum (Chordata) and Kingdom (Animalia).

117. Answer (2)

Hint: Telophase is the final stage of karyokinesis.

Sol.: Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

118. Answer (3)

Hint: Crossing over is completed by the end of third substage of prophase I.

Sol.: Crossing over takes place in pachytene.

119. Answer (3)

Hint: Poales is the order to which wheat belongs.

Sol.: Plant families like Convolvulaceae and Solanaceae are included in the order Polymoniales.

120. Answer (3)

Hint: Muscidae belongs to Diptera.

Sol.: The animal order, Carnivora, includes families like Felidae and Canidae.

121. Answer (4)

Hint: The sequence of events by which a cell duplicates its genome, synthesising the other constituents of the cell and eventually divides into two daughter cells is termed as cell cycle.

Sol.: The period of cell cycle varies from organism to organism and from cell to cell of an organism.

122. Answer (1)

Hint: It occurs in the final stage of meiotic prophase I.

Sol.: By the end of diakinesis, the nucleolus disappears and the nuclear envelope also breaks down.

123. Answer (3)

Hint: *Panthera leo* is the name of lion.

Sol.: The correct scientific name of leopard is *Panthera pardus*.

124. Answer (2)

Hint: ICZN provides the naming principles for animals.

Sol.: Animals are eukaryotes.

125. Answer (2)

Hint: Cell plate represents the middle lamella between the walls of two adjacent cells.

Sol.: The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. Chiasmata is the site of crossover. During prophase, the two asters together with spindle fibres forms mitotic apparatus.

126. Answer (4)

Hint: Only a single cycle of DNA replication is observed during meiosis.

Sol.: There is no replication of DNA during interkinesis.

127. Answer (4)

Hint: The scientific name of the organism is constituted of specific epithet and generic name.

Sol.: Scientific name gives description about species and genus of the organism along with the author name.

128. Answer (4)

Hint: Meiosis involves only a single cycle of DNA replication.

Sol.: Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S-phase. Cell division does not stop with the formation of the mature organism but continues throughout its life cycle.

129. Answer (4)

Hint: It forms at the end of meiosis II.

Sol.: Cytokinesis that follows telophase II results in the formation of a tetrad of cells.

130. Answer (3)

Hint: *Solanum nigrum* is the scientific name of makoi.

Sol.: *familiaris* belongs to genus *Canis*.

131. Answer (3)

Sol.: Biological names are generally in Latin.

132. Answer (1)

Hint: Cell division restores the nucleo-cytoplasmic ratio.

Sol.: Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.

133. Answer (1)

Hint: The life cycle begins with zygote.

Sol.: Any sexually reproducing organism starts its life cycle from a single-celled zygote (diploid).

134. Answer (3)

Hint: M phase is the most dramatic period of the cell cycle

Sol.: M phase involves major reorganisation of virtually all components of the cell.

135. Answer (3)

Hint: The mentioned phase is diplotene

Sol.: Diplotene shows the appearance of chiasmata.

[ZOOLOGY]

136. Answer (4)

Hint: TV is approximately 500 mL.

Sol.: EC = TV + ERV = 1500 – 1600 mL.

So, value of expiratory capacity is almost equal to three times the tidal volume (TV).

Inspiratory capacity = 3000 – 3500 mL

IRV = 2500 – 3000 mL

RV = 1100 – 1200 mL

137. Answer (3)

Hint: Eliminate the one which is released out.

Sol.: Oxygen (O_2) is utilized by the organisms to indirectly breakdown simple molecules like glucose, amino acids, fatty acids, *etc*., to derive energy to perform various activities. CO_2 is released during catabolic reactions.

138. Answer (4)

Hint: Partial pressure of gases affects diffusion.

Sol.: Total pressure of a gaseous mixture is the sum of partial pressures of different gaseous constituents present in the gaseous mixture and remains the same for all the gases. So, total pressure of a gaseous mixture is not responsible for their differential diffusion. Partial pressure of different gases is responsible for their different rates of diffusion.

139. Answer (1)

Hint: Division of respiratory tract after bronchi.

Sol.: Only trachea, bronchi, and initial bronchioles of the conducting part of human's respiratory system are supported by incomplete cartilaginous rings. Nasal passage, pharynx, and terminal bronchioles are not supported by incomplete cartilaginous rings.

140. Answer (2)

Hint: Diffusion of gases depends upon their solubility.

Sol.: As the solubility of CO_2 is 20–25 times higher than that of O_2, the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2.

141. Answer (4)

Hint: 1 $mm^{-3} = 10^{-3}$ mL.

Sol.: A healthy individual has 12–16 gms of haemoglobin in every 100 mL (1 decilitre) of blood. 5.5 million RBCs are present in 0.001 mL of blood and 55 million RBCs are present in 0.01 mL of blood.

142. Answer (3)

Hint: These cells are devoid of nucleus.

Sol.: In most of the mammals including humans, RBCs are devoid of nucleus and are maximum in number among other formed elements. Neutrophils are maximum in number among WBCs and granulocytes, whereas lymphocytes are maximum in number among agranulocytes.

143. Answer (4)

Hint: Rh factor of foetal RBCs is immunogen for mother.

Sol.: Erythroblastosis foetalis, a condition in which mother starts to produce anti-Rh antibodies against Rh factor of foetal blood as mother is Rh-ve, so, anti-Rh antibodies are administered to mother to destroy Rh antigen immediately to prevent mother's immune system to activate against Rh-antigen.

144. Answer (3)

Hint: Lymph is also a fluid connective tissue circulating in lymphatic system.

Sol.: Some water along with water soluble substances which filter out in the region of capillary bed is called tissue fluid. It is devoid of most of the formed elements, and few larger proteins present in blood. It is responsible for transportation of nutrients and is also a carrier for hormones. It contains lymphocytes but lacks RBCs.

145. Answer (1)

Hint: RBCs undergo clumping due to transfusion of mismatched blood.

Sol.: Thrombokinase is produced by cascade processes which is responsible for conversion of prothrombin into thrombin. Clot is mainly formed by a network of threads called fibrins in which formed elements of blood are trapped.

146. Answer (3)

Hint: Additional muscles in the abdomen help in forceful breathing.

Sol.: Normal breathing is performed by contraction and relaxation of diaphragm and external inter-costal muscles. Contraction of internal inter-costal muscles and abdominal muscles help in forceful expiration. Muscles in the abdomen have ability to increase the strength of forceful inspiration as well as expiration.

147. Answer (4)

Hint: Factors are favourable for diffusion of CO_2 in opposite direction than O_2.

Sol.: Rate of diffusion of gases depends upon partial pressure of gases, thickness of diffusion membrane, and solubility of gases. All these factors facilitate diffusion of O_2 from alveoli to deoxygenated blood and from oxygenated blood to tissues, whereas diffusion of CO_2 is facilitated from tissues to deoxygenated blood and from deoxygenated blood to alveoli.

148. Answer (2)

Hint: Exclude the function of diffusion membrane.

Sol.: Respiratory system of human beings is differentiated into two parts: conducting part and exchange part. The parts starting with the external nostrils up to the terminal bronchioles constitute the conducting part, whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system. All mentioned functions are associated with conducting part, whereas exchange part provides surface for gaseous exchange.

149. Answer (2)

Hint: 100 mL of oxygenated blood delivers 5 mL of O_2 to tissues under normal physiological conditions.

Sol.: Blood pumped out by left ventricle in each cardiac cycle = Stroke volume = 70 mL

Blood pumped out by left ventricle per minute = Cardiac Output = HR × SV = 72 × 70 $\simeq$ 5000 mL
Because 100 mL of oxygenated blood delivers only 5 mL of O_2 to tissue, so oxygen delivered to tissue per minute = $\frac{5}{100} \times 5000 = 250$ mL.

150. Answer (1)

Hint: Overall decrease in intra-pulmonary pressure causes inspiration.

Sol.: Contraction of diaphragm increases the volume of thoracic chamber in antero-posterior axis and contraction of external inter-costal muscles increases volume of thoracic chamber in dorso-ventral axis that leads to increase in volume of lungs. As a result, intra-pulmonary pressure decreases below atmospheric pressure to create a suction pull to rush air into lung alveoli from atmosphere *i.e.*, inspiration.

151. Answer (2)

Hint: Animals with nephridia for excretion.

Sol.: The closed circulatory system with a heart to pump blood in blood vessels appeared for the first time in members of the phylum Annelida, and open circulatory system appeared for the first time in members of the phylum Arthropoda.

152. Answer (4)

Hint: Think about pulmonary artery.

Sol.: Arteries run away from the heart and usually carry oxygenated blood with few exceptions, *e.g.*, pulmonary arteries which carry deoxygenated blood. On the other hand, veins run towards the heart and usually carry deoxygenated blood with few exceptions, *e.g.*, pulmonary veins which carry oxygenated blood from lungs to left auricle.

153. Answer (4)

Hint: Fishes exhibit single circulation.

Sol.: Animal 'A' is a fish and 'B' is an amphibian. In fishes, deoxygenated blood received by atrium is pumped to gills for oxygenation and blood passes only once through heart, called single circulation. In amphibians, deoxygenated blood is received by right atrium and oxygenated blood is received by left atrium. Blood from both atria reaches a single ventricle and this mixed blood is pumped to

systems of the body; such type of circulation is called incomplete double circulation.

154. Answer (4)

Hint: Ratio of diastolic to systolic blood pressure is 80 : 120 = 2 : 3.

Sol.: pO_2 at which haemoglobin is 75% saturated with O_2 = 40 mm Hg

pO_2 at which haemoglobin is 90% saturated with O_2 is about 60 mm Hg

Ratio of these two values of pO_2 = 40 : 60 = 2 : 3.

Per cent of lymphocytes in the total WBCs = 20 – 25%.

Per cent of CO_2 transported in the form of carbamino-haemoglobin = 20 – 25%.

Transportation of CO_2 dissolved in plasma = 7%.
Transport of CO_2 as bicarbonate in plasma = 70%.

So, per cent of CO_2 transported in plasma as bicarbonate is 10 times to the CO_2 transported in dissolved state in plasma.

155. Answer (1)

Hint: Semilunar valves close when pressure within ventricles decreases.

Sol.: During joint diastole, semilunar valves close and AV valves open and blood freely moves from vena cava to right atrium and from atria to ventricles. Closure of AV valves occurs during ventricular systole. Only AV valves remain open during joint diastole.

156. Answer (1)

Hint: Cardiac output = Heart rate × Stroke volume.

Sol.: Stroke volume = End diastole volume – End systole volume

= [140 – 60] = 80 mL

Cardiac output = Heart rate × Stroke volume

$= \frac{50}{\text{min}} \times 80 \text{ mL} = 4000 \text{ mL/min} = 4 \text{ L/min}$

157. Answer (2)

Hint: Capillaries are connecting link between arterial system and venous system.

Sol.:

Coronary system	–	Associated with blood supply to heart
Hepatic portal vein	–	Drains blood from intestine to liver
Capillary	–	A link between arteriole and venule
Vena cava	–	Largest vein that opens into right atrium of heart

158. Answer (1)

Hint: T-P gap of ECG represents joint diastole.

Sol.: TP gap of ECG represents joint diastole during which heart muscles remain electrically silent. Eosinophils release histaminase which degrades histamine released from basophils and mast cells.

159. Answer (3)

Hint: Elasticity is the property shown by elastic fibres and muscle fibres.

Sol.: Tunica media is a middle layer of blood vessels with smooth muscle and elastic fibres. Both muscle fibres and elastic fibres of connective tissue exhibit a property called elasticity.

160. Answer (2)

Hint: Constitute minimum per cent of the total WBCs.

Sol.: Basophils and mast cells release histamine which participate in inflammatory reactions of the body. Cortisol is responsible for anti-inflammatory reactions of the body.

161. Answer (4)

Hint: It is situated close to atrio-ventricular septum.

Sol.: SAN (Sino-Atrial Node) is a patch of nodal tissue present in the right upper corner of right atrium and AVN (Atrio-Ventricular Node) is another patch of nodal tissue situated in lower left corner of the right atrium.

162. Answer (4)

Hint: Function of sympathetic and parasympathetic neural system is antagonistic to each other.

Sol.: A specialised neural centre in the medulla oblongata can moderate the cardiac functions through autonomic nervous system. Neural signals through the sympathetic nerve fibres can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output. On the other hand, parasympathetic neural signals decrease the rate of heart beat, speed of conduction of action potentials, the strength of ventricular systole and thereby the cardiac output. Adrenal medullary hormones also perform the function of sympathetic neural system.

163. Answer (3)

Hint: The unique shape of mammalian RBCs relates to their function of transporting O_2.

Sol.: Mammalian RBCs are superior than RBCs of other vertebrates because:

(i) Biconcave shape of mammalian RBCs provides more surface area for diffusion of gases.

(ii) Absence of nucleus, mitochondria, *etc.*, provide large volume for storage of respiratory chromoprotein called haemoglobin.

164. Answer (4)

Hint: Universal recipient individuals lack antibodies against all blood group antigens.

Sol.: AB+ve blood group individuals are considered as universal recipient because antibodies against blood group antigens are not present in their blood plasma. Serum contains antibodies but lacks clotting factors. Blood pressure during ventricular systole is maximum in aorta due to its minimum total cross-section area and minimum in capillaries due to their maximum total cross-section area.

165. Answer (3)

Hint: Exclude the pacemaker of heart.

Sol.: SA node generates impulses at maximum rate and is considered as pacemaker of heart. AV node has minimum conduction velocity so that it synchronises atrial and ventricular systole in a synchronised order. So, AV node is considered as pace setter of the heart. AV Bundle transmits impulses from AVN of atria to the ventricles and electrically connects atria to ventricles for transmission of impulses.

166. Answer (4)

Hint: Intra-pleural pressure decreases on increasing volume of thoracic cavity.

Sol.: Normal inspiration takes place due to contraction of diaphragm that leads to increase in diameter/length of thoracic cavity in antero-posterior axis. Due to downward movement of diaphragm, the volume of thoracic cavity and thus intra-pulmonary volume increases, and that of abdominal cavity decreases, increasing intra-abdominal pressure.

167. Answer (2)

Hint: Chemosensitive area in peripheral and central chemoreceptors are sensitive to H^+ and CO_2.

Sol.: Change in concentration of H^+ and CO_2 in arterial oxygenated blood affects chemoreceptors of aortic arch and carotid body. Similarly, change in concentration of H^+ and CO_2 in CSF affects the chemosensitive area present in medulla oblongata. Variation in arterial Na^+ concentration does not affect respiration.

168. Answer (2)

Hint: Decrease in CO_2 shifts oxygen dissociation curve to left side.

Sol.: The factors which favour oxygenation of haemoglobin to increase its percentage saturation are:

(i) Decrease in pCO_2

(ii) Decrease in temperature

(iii) Increase in pH

(iv) Decrease in H^+ concentration

(v) Decrease in 2,3-BPG

169. Answer (3)

Hint: The dome-shaped structure forms floor of thoracic cavity.

Sol.: The chamber formed dorsally by the vertebral column, ventrally by sternum, laterally by ribs, and on the lower side by dome-shaped diaphragm is called thoracic cavity.

170. Answer (2)

Hint: Atherosclerosis leads to coronary artery disease.

Sol.: Asthma is an allergic disease characterised by wheezing during respiration due to inflammation of bronchi and bronchioles. In emphysema, respiratory surface decreases due to damage of alveolar walls. Lung fibrosis occurs in occupational lung diseases such as asbestosis. Coronary artery disease is associated with taking fat rich diet in which deposition of fat, cholesterol, calcium in lining of blood vessels leads to narrowing of their lumen.

171. Answer (1)

Hint: Stimulation of chemoreceptors in carotid bodies increases breathing rate.

Sol.: Stimulation of chemoreceptors in carotid bodies increases breathing rate. Injection of drug that activates chemoreceptors in carotid bodies would be expected to increase breathing rate and thus increase in the pO_2 of arterial blood. In CO is poisoning, CO tightly bound to haemoglobin so that dissociation of oxyhaemoglobin as well as binding of O_2 with haemoglobin becomes poor.

172. Answer (3)

Hint: Animals with tissue level of organisation.

Sol.: Simple organisms like sponges and coelenterates circulate water from their surroundings through their body cavities to facilitate the cells to exchange the substances.

173. Answer (2)

Hint: Cells which constitute maximum per cent of the total WBCs.

Sol.: Neutrophils are polymorphonuclear leucocytes which constitute 60-65% of the total WBCs. Maximum per cent of blood corpuscles is formed by RBCs.

174. Answer (3)

Hint: Number of QRS complexes in a given time represents heart beat rate.

Sol.: To obtain a standard ECG, a patient is connected to machine with three electrical leads (one to each wrist and to the left ankle). Letters of ECG from P to T corresponds to a specific electrical activity of heart. By counting the number of QRS complexes that occur in a given time period, one can determine heart beat rate of an individual.

175. Answer (3)

Hint: VC = TLC – RV.

Sol.: Vital capacity = TV + IRV + ERV
= 500 + (2500-3000) mL + (1000-1100) mL
= 4100 – 4600 mL.

176. Answer (2)

Hint: Intra-pleural pressure is less than atmospheric pressure

Sol.: After forceful expiration, residual volume left in lungs, cannot be expelled out due to negative intra-pleural pressure which pulls lung walls to outside. So, it prevents elastic recoiling of lungs and thus prevents alveoli to collapse.

177. Answer (2)

Hint: Haemoglobin is saturated upto 50% when pO_2 is less than 40 mmHg.

Sol.: P_{50} value of haemoglobin is about 28 mmHg in the range of 25-30 mmHg. Increase in P_{50} value decreases affinity of Hb for O_2 and *vice versa*.

178. Answer (2)

Hint: Left side of heart pumps blood to systems of the body.

Sol.: Right atrium and left ventricle are included in systemic circulation as former receives deoxygenated blood from systems of the body and left ventricle pumps oxygenated blood to systems of the body. Similarly, right ventricle and left atrium are parts of pulmonary circulation because right ventricle pumps deoxygenated blood to lungs and left atrium receives oxygenated blood from lungs.

179. Answer (2)

Hint: In cardiac arrest, heart completely stops its function.

Sol.:

Angina pectoris	–	Acute chest pain due to low supply of O_2 to heart muscles.
Heart failure	–	Heart is not pumping blood effectively as needs of the body.
Cardiac arrest	–	Heart stops beating.
Heart attack	–	Heart muscles are suddenly damaged due to low O_2 supply.
Hypertension		BP > 140/90 of a patient on repeated checks.

180. Answer (4)

Hint: 'Dub' sound is produced due to closure of semilunar valves.

Sol.: 100% filling of ventricles occurs at the end of ventricular diastole after atrial systole. Heart sound 'Lub' is produced at the beginning of ventricular systole due to closure of AV valves whereas 'dub' sound is produced at the beginning of ventricular diastole due to closure of semilunar valves.

❑ ❑ ❑

All India Aakash Test Series for NEET-2028

TEST - 2 (Code-D)

Click here for Code-C Sol.

Test Date : 25/10/2026

ANSWERS

1. (1)	37. (2)	73. (3)	109. (2)	145. (1)
2. (1)	38. (2)	74. (3)	110. (2)	146. (3)
3. (2)	39. (4)	75. (4)	111. (2)	147. (4)
4. (3)	40. (3)	76. (2)	112. (1)	148. (2)
5. (1)	41. (3)	77. (4)	113. (4)	149. (2)
6. (2)	42. (2)	78. (2)	114. (4)	150. (1)
7. (3)	43. (1)	79. (2)	115. (2)	151. (2)
8. (1)	44. (1)	80. (2)	116. (1)	152. (4)
9. (2)	45. (3)	81. (4)	117. (4)	153. (4)
10. (3)	46. (2)	82. (3)	118. (2)	154. (4)
11. (3)	47. (2)	83. (1)	119. (4)	155. (1)
12. (1)	48. (2)	84. (4)	120. (3)	156. (1)
13. (1)	49. (3)	85. (1)	121. (2)	157. (2)
14. (3)	50. (1)	86. (4)	122. (1)	158. (1)
15. (3)	51. (3)	87. (3)	123. (1)	159. (3)
16. (2)	52. (2)	88. (3)	124. (4)	160. (2)
17. (2)	53. (2)	89. (4)	125. (2)	161. (4)
18. (4)	54. (2)	90. (1)	126. (2)	162. (4)
19. (3)	55. (1)	91. (3)	127. (1)	163. (3)
20. (4)	56. (2)	92. (3)	128. (2)	164. (4)
21. (1)	57. (2)	93. (1)	129. (1)	165. (3)
22. (4)	58. (4)	94. (1)	130. (1)	166. (4)
23. (4)	59. (1)	95. (3)	131. (2)	167. (2)
24. (2)	60. (2)	96. (3)	132. (1)	168. (2)
25. (3)	61. (4)	97. (4)	133. (4)	169. (3)
26. (3)	62. (2)	98. (4)	134. (1)	170. (2)
27. (4)	63. (3)	99. (4)	135. (3)	171. (1)
28. (1)	64. (3)	100. (4)	136. (4)	172. (3)
29. (4)	65. (4)	101. (2)	137. (3)	173. (2)
30. (2)	66. (2)	102. (2)	138. (4)	174. (3)
31. (2)	67. (1)	103. (3)	139. (1)	175. (3)
32. (1)	68. (3)	104. (1)	140. (2)	176. (2)
33. (4)	69. (1)	105. (4)	141. (4)	177. (2)
34. (4)	70. (1)	106. (3)	142. (3)	178. (2)
35. (1)	71. (1)	107. (3)	143. (4)	179. (2)
36. (1)	72. (3)	108. (3)	144. (3)	180. (4)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (1)

Hint & Sol.: $\sin(\pi t) = \frac{x}{a}$ ...(i)

$\cos(\pi t) = \frac{y}{b}$ ...(ii)

On squaring (i) and (ii), and then adding,

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \sin^2(\pi t) + \cos^2(\pi t)$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2} = 1 \quad (\because\ b = a)$

$\Rightarrow x^2 + y^2 = a^2$, is equation of circle.

2. Answer (1)

Hint: Split $\overrightarrow{OB}$ in x-y rectangular components.

Sol.: $\vec{R} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$

$= r\hat{i} + r\cos 45^\circ \hat{i} + r\sin 45^\circ \hat{j} + r\hat{j}$

$= \left(r + \frac{r}{\sqrt{2}}\right)\hat{i} + \left(r + \frac{r}{\sqrt{2}}\right)\hat{j}$

$= (\sqrt{2}r + r)(\cos 45^\circ \hat{i} + \sin 45^\circ \hat{j})$

$\vec{R} = \left(\sqrt{2}r + r\right)$ along $\overrightarrow{OB}$

3. Answer (2)

Hint: Velocity, $\vec{v} = \frac{d\vec{r}}{dt}$

Sol.: $\vec{v} = 3\hat{i} + 6\hat{j}$

$\Rightarrow$ Velocity along y-axis = 6 unit.

4. Answer (3)

Hint: Compare given equation $y = \frac{x}{\sqrt{3}} - \frac{gx^2}{6}$ with

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

Sol.: $\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = 30^\circ$

$2u^2\cos^2\theta = 6$

$u^2 \times \frac{3}{4} = 3$

$u = 2$ m/s

5. Answer (1)

Hint: For collision, vertical component of velocity of both particles should be same.

Sol.: $v_A \sin\theta = v_B$

$20\sin\theta = 10$

$\sin\theta = \frac{1}{2}$

$\theta = 30^\circ$

$x = v_A \cos 30^\circ \times t$

$= 20 \times \frac{\sqrt{3}}{2} \times \frac{1}{2}$

$= 5\sqrt{3}$ m

6. Answer (2)

Hint: For $\vec{a} \parallel \vec{b}$, we have $\frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z}$

Sol.: $\vec{a} = 2\hat{i} + 3\hat{j} - 2\hat{k}$

$\vec{b} = \hat{i} + \lambda\hat{j} + \mu\hat{k}$

$\therefore\ 2 = \frac{3}{\lambda} = \frac{-2}{\mu}$

$\Rightarrow \frac{\lambda}{\mu} = -\frac{3}{2}$

7. Answer (3)

Hint: Net acceleration $a = \sqrt{a_x^2 + a_y^2}$

Sol.: $s = ut + \frac{1}{2}at^2$, along x direction gives

$6 = \frac{1}{2}a_x(2)^2 \Rightarrow a_x = 3$ m/s^2

$s = ut + \frac{1}{2}at^2$, along y-direction gives

$8 = \frac{1}{2}a_y(2)^2 \Rightarrow a_y = 4$ m/s^2

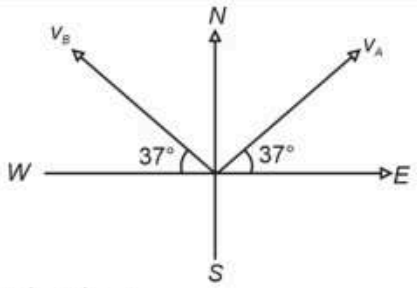
$a = \sqrt{3^2 + 4^2}$

$= 5$ m/s^2

8. Answer (1)

Hint: Use relative velocity $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$

Sol.:



$\vec{v}_A = (8\hat{i} + 6\hat{j})$ m/s

$\vec{v}_B = (-8\hat{i} + 6\hat{j})$ m/s

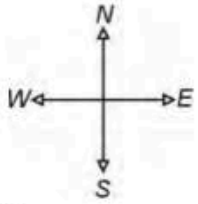
$\vec{v}_{AB} = 8\hat{i} + 6\hat{j} + 8\hat{i} - 6\hat{j}$

$= 16\hat{i}$ m/s

9. Answer (2)

Hint: On multiplying a vector by (–1), the direction of the vector becomes opposite.

Sol.:



$\vec{d} = 5\hat{j}$ km

$\vec{d}_1 = -2 \times 5\hat{j}$

$= -10\hat{j}$ km (along south)

10. Answer (3)

Hint & Sol.: Magnitude of velocity and magnitude of acceleration are both constant in uniform circular motion, so change in magnitude will be zero.

11. Answer (3)

Hint & Sol.: x-component, $R_x = R\cos\theta$

while y – component, $R_y = R\sin\theta$

$R_x \times R_y = R^2 \sin\theta \times \cos\theta$

$= \dfrac{R^2 \sin 2\theta}{2}$

12. Answer (1)

Hint: Resultant $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Sol.: $R = |\vec{C}|$

$|\vec{C}| = \sqrt{A^2 + B^2 + 2AB\cos 180°}$

$|\vec{C}| = \sqrt{A^2 + B^2 - 2AB}$

$= A - B$

$= |\vec{A}| - |\vec{B}|$

13. Answer (1)

Hint: y-component of velocity is given by $v_y = \dfrac{dy}{dt}$

Sol.: $y = 2x^2$

$y = 2t^4$

$\dfrac{dy}{dt} = 8t^3$

$v_y = 8 \times \left(\dfrac{1}{2}\right)^3$

$= 1$ m/s

14. Answer (3)

Hint: Horizontal component of the velocity of object will remain same.

Sol.: $40\cos 53° = v\cos 37°$

$40 \times \dfrac{3}{5} = v \times \dfrac{4}{5}$

$v = 30$ m/s

$v_y = v\sin 37°$

$= 30 \times \dfrac{3}{5}$

$= 18$ m/s

15. Answer (3)

Hint & Sol.: In oblique projectile motion of an object, acceleration remains constant but speed continuously changes.

16. Answer (2)

Hint: Time of flight $T = \dfrac{2u_y}{g}$, Range $R = u_x T$

Sol.: $\tan\theta = \dfrac{u_y}{u_x}$

$\tan\theta = \dfrac{4}{3}$

$\theta = 53°$

From vertical, $90° - \theta = 37°$

$T = \dfrac{2 \times 40}{10} = 8$ s

$R = 30 \times 8 = 240$ m

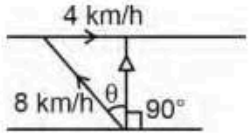
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$$H = \frac{u_y^2}{2g} = \frac{(40)^2}{20} = 80 \text{ m}$$

17. Answer (2)

Hint: To travel with minimum drift (along shortest path), angle made by boat with river flow should be obtuse.

Sol.:



$$\sin\theta = \frac{4}{8} \Rightarrow \theta = 30°$$

$\therefore$ Required angle = 90° + 30°

= 120°

18. Answer (4)

Hint: Unit vector $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

Sol.: Required vector $\vec{b} = b\hat{A}$

$$\hat{A} = \frac{2\hat{i} + 2\hat{j}}{\sqrt{2^2 + 2^2}}$$

$$\hat{A} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

$$\vec{b} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)4\sqrt{2} = \left(4\hat{i} + 4\hat{j}\right)$$

19. Answer (3)

Hint: Use resultant $R = \sqrt{a^2 + b^2 + 2ab\cos\theta}$

Sol.: $\sqrt{10}p = \sqrt{(2p)^2 + \left(\sqrt{2}p\right)^2 + 2\times 2p\left(\sqrt{2}p\right)\cos\theta}$

$10p^2 = 6p^2 + 4\sqrt{2}p^2\cos\theta$

$4p^2 = 4\sqrt{2}p^2\cos\theta$

$$\cos\theta = \frac{1}{\sqrt{2}}$$

$\theta = 45°$

20. Answer (4)

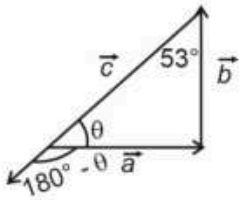
Hint & Sol.: Direction of angular velocity of object moving in $x - y$ plane on a circular path, may be along positive or negative z-direction depending upon sense of rotation.

21. Answer (1)

Hint & Sol.: When angle between velocity and acceleration is obtuse, then one component of acceleration (tangential acceleration) decreases the magnitude of velocity while other component of acceleration (normal acceleration) changes the direction of velocity.

22. Answer (4)

Hint & Sol.:



In this case, $\theta = 37°$

Required angle = 180° – θ = 180 – 37°

= 143°

23. Answer (4)

Hint: Radius of curvature at maximum height of projectile,

$$R = \frac{(u\cos\theta)^2}{g}$$

Sol.: $$\frac{R}{H} = \frac{u^2\cos^2\theta}{g\times\frac{u^2\sin^2\theta}{2g}}$$

$$= 2\cot^2\theta = 2\times\left(\frac{1}{\sqrt{3}}\right)^2$$

$$= \frac{2}{3}$$

24. Answer (2)

Hint: Displacement $d = \sqrt{H^2 + \left(\frac{R}{2}\right)^2}$

Sol.: $$d = \sqrt{\left(\frac{u^2\sin^2\theta}{2g}\right)^2 + \left(\frac{u^2\sin 2\theta}{2g}\right)^2}$$

$$d = \frac{u^2}{2g}\sqrt{\sin^4\theta + \sin^2 2\theta}$$

$$d = \frac{u^2}{2g}\sqrt{\frac{1}{16} + \frac{3}{4}}$$

$$= \frac{u^2}{8g}\sqrt{13}$$

25. Answer (3)

Hint: Range $R = (u\cos\theta)\,T$

Sol.: $H_1 = H_2$

$u_1\sin\theta_1 = u_2\sin\theta_2 \Rightarrow T_1 = T_2$

$\sin\theta_1 > \sin\theta_2$ $(\because \theta_1 > \theta_2)$

$\Rightarrow$ $u_1 < u_2$ and $\cos\theta_1 < \cos\theta_2$

$R_1 = (u_1\cos\theta_1)\, T_1$

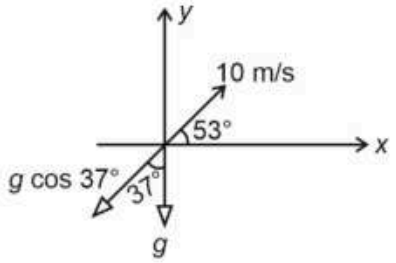
$R_2 = (u_2\cos\theta_2)\, T_2$

$\therefore R_1 < R_2$

26. Answer (3)

Hint: The rate of change of speed is equal to tangential acceleration of object.

Sol.:



$a_T = g\cos 37°$

$= 8\ \text{m/s}^2$

27. Answer (4)

Hint & Sol.: Displacement, velocity and acceleration all of them are vector quantities.

28. Answer (1)

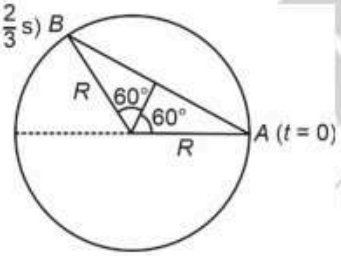
Hint & Sol.: In oblique projectile motion, time of upward journey is equal to time taken for downward journey. At maximum height, speed will be minimum.

29. Answer (4)

Hint: Angular displacement = $\omega \times t$

$= \dfrac{2\pi}{3}\text{ rad} = 120°$

Sol.:



Displacement = $2R\cos 30°$

$= 2R \times \dfrac{\sqrt{3}}{2}$

$= \sqrt{3}R$

30. Answer (2)

Hint & Sol.: In uniform circular motion, direction of velocity and direction of centripetal acceleration will both continuously change, but magnitude and direction of angular velocity will both remain same.

31. Answer (2)

Hint: Time taken in quicker path, $t = \dfrac{d}{V_{BR}}$

Sol.: Time taken in shortest path,

$T = \dfrac{d}{\sqrt{V_{BR}^2 - V_R^2}}$

$5 + t = T$

$5 + \dfrac{d}{V_{BR}} = \dfrac{d}{\sqrt{V_{BR}^2 - V_R^2}}$

$\dfrac{d}{6} - \dfrac{d}{10} = 5$

$\dfrac{5d - 3d}{30} = 5$

$d = 75\ \text{m}$

32. Answer (1)

Hint: The relation between x and y gives the equation of trajectory.

Sol.: $\vec{a} = \left(2\hat{i} + 3\hat{j}\right)\text{m/s}^2$

Writing $s = ut + \dfrac{1}{2}at^2$ along x-axis

$x = \dfrac{1}{2}(2)t^2$

$x = t^2$ ...(1)

Writing $s = \left(ut + \dfrac{1}{2}at^2\right)$ along y-axis

$y = \dfrac{3}{2}t^2$ ...(2)

$y = \dfrac{3}{2}x$

$2y = 3x$

33. Answer (4)

Hint & Sol.:

In any triangle, length of the third side is always less than the sum of the lengths of other two sides but greater than their difference.

34. Answer (4)

Hint: Acceleration of object w.r.t, observer $\vec{a}_{PO} = -2g\hat{j}$, with $\hat{j}$ along vertically upward direction.

Sol.: $x = v_0 t$

$$y = -\frac{1}{2}2gt^2$$

$y = \frac{-gx^2}{v_0^2}$ is equation of parabola.

35. Answer (1)

Hint: Resultant $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Sol.: For parallel vectors, $\theta = 0°$

$\Rightarrow R = A + B$

For anti-parallel vectors, $\theta = 180°$

$\Rightarrow R = A - B$

For $\theta = 60°$,

$R = \sqrt{A^2 + B^2 + AB}$

For $\theta = 120°$,

$R = \sqrt{A^2 + B^2 - AB}$

$\therefore R_a > R_c > R_d > R_b$

36. Answer (1)

Hint & Sol.: Magnitude of horizontal component and vertical component of velocity, are same at both point of projection and landing point. Therefore, speed at both the points will be same.

37. Answer (2)

Hint: Compare equation of trajectory $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$ with given equation.

Sol.: On comparing,

$\tan\theta = \sqrt{3} \Rightarrow \theta = 60°$

$u^2\cos^2\theta = 1 \Rightarrow u = 2$ m/s

$T = \frac{2u\sin\theta}{g}$

$= \frac{2\times 2\sin 60°}{10}$

$= \frac{\sqrt{3}}{5}$ s

38. Answer (2)

Hint: Use range $R = \frac{u^2\sin 2\theta}{g}$

Sol.: $\frac{\left(\frac{v}{2}\right)^2 \sin 90°}{g} = \frac{v^2\sin 2\theta}{g}$

$\theta = \frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$

39. Answer (4)

Hint: Horizontal component of velocity will remain same.

Sol.: $v_y = u_y - gt$

$v_y = 40 - 10t$, $v_x = 10$ m/s

$\tan 45° = \left(\frac{v_y}{v_x}\right)$

$\frac{40 - 10t}{10} = 1$

$10t = 40 - 10$

$t = 3$ s

40. Answer (3)

Hint: Use $S = ut + \frac{1}{2}at^2$ in vector form

Sol.: $\vec{x} - \vec{x}_0 = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$\vec{x} - \vec{x}_0 = \frac{1}{2}(2\hat{i} + 3\hat{k})(2)^2$

$\vec{x} - \vec{x}_0 = 4\hat{i} + 6\hat{k}$

$\vec{x} - (\hat{j} + 3\hat{k}) = 4\hat{i} + 6\hat{k}$

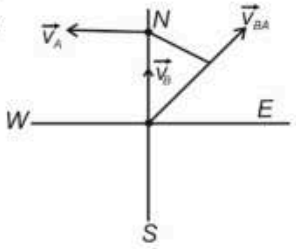
$\vec{x} = (4\hat{i} + \hat{j} + 9\hat{k})$ m

41. Answer (3)

Hint: Use concept of relative motion.

$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A}$

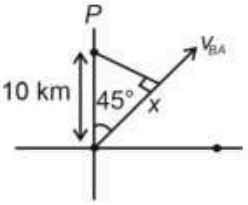
Sol.:



$\overrightarrow{v_{BA}} = \overrightarrow{v_B} - \overrightarrow{v_A}$

$\overrightarrow{v_{BA}} = (20\hat{i} + 20\hat{j})$ km/h

$|\overrightarrow{v_{BA}}| = 20\sqrt{2}$ km/h



$x = 5\sqrt{2}$ km

$t = \frac{x}{|v_{BA}|} = \frac{5\sqrt{2}}{20\sqrt{2}}$

$= \frac{1}{4}$h = 15 min

42. Answer (2)

Hint: Use $\cos\theta = \frac{d_x}{|\vec{d}|}$

Sol.: $\vec{d} = (3-1)\hat{i} + (5-2)\hat{j} + (2-1)\hat{k}$

$|\vec{d}| = \sqrt{(2^2 + 3^2 + 1^2)}$

$= \sqrt{14}$ units

$d_x = 2$ units

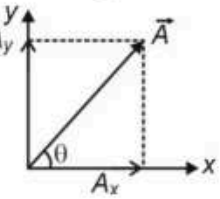
$\cos\theta = \left(\frac{2}{\sqrt{14}}\right)$

$\Rightarrow \theta = \cos^{-1}\left(\frac{2}{\sqrt{14}}\right)$

43. Answer (1)

Hint: Rectangular component of vector depends on angle θ made by vector with x-axis or y-axis.

Sol.:



$A_x = A\cos\theta$

$A_y = A\sin\theta$

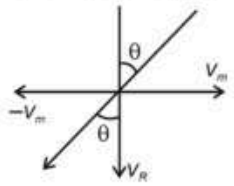
On changing value of θ, A_x and A_y will change.

A vector can be split into infinite number of components.

44. Answer (1)

Hint: Use the concept of relative motion

Sol.:



$\overrightarrow{v_{Rm}} = \overrightarrow{v_R} - \overrightarrow{v_m}$

$\tan\theta = \frac{v_m}{v_R}$

$\tan\theta = \frac{8}{6} \Rightarrow \theta = 53°$

45. Answer (3)

Hint: Use $h = \frac{u^2\sin^2\theta}{2g}$ and time of flight $T = \frac{2u\sin\theta}{g}$

Sol.: $T^2 = \frac{4u^2\sin^2\theta}{g^2}$

$\therefore h = \frac{u^2\sin^2\theta}{2g} = \frac{gT^2}{8}$

$= \frac{10\times 36}{8}$

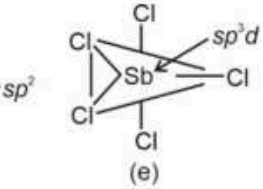
= 45 m

[CHEMISTRY]

46. Answer (2)

Hint: 120° bond angle is observed in species having sp^2 or sp^3d hybridisation

Sol.:



(a)

(e)

Both (a) and (e) contain 120° bond angles.

47. Answer (2)

Hint: Element (X) is aluminium

Sol.: X_2Y_3 type of compound will be formed if Y belongs to oxygen family. Hence Y is oxygen having electronic configuration $1s^2 2s^2 2p^4$.

48. Answer (2)

Hint: If a system absorb energy, then the process is called endothermic.

Sol.: To ionise an atom, energy is always required to remove an electron from the atom.

49. Answer (3)

Hint: Isoelectronic species contain same number of electrons.

Sol.: Cl^-, Ar, K^+ and Sc^{3+} contain 18 electrons each, hence these are isoelectronic species.

50. Answer (1)

Hint: For a given anion, smaller is the size of cation and greater the charge, greater will be the covalent character of ionic bond.

Sol.: • Correct order of ionic size :

$$Na^+ > Mg^{2+} > Al^{3+}$$

- Correct order of covalent character

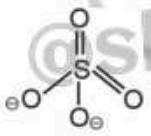
  $AlCl_3 > MgCl_2 > NaCl$

51. Answer (3)

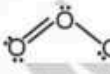
Hint: Formal charge

$$= \left[\begin{array}{l}\text{Total number}\\ \text{of valence}\\ \text{electrons in}\\ \text{the free atom}\end{array}\right] - \left[\begin{array}{l}\text{Total number}\\ \text{of non bonding}\\ \text{electrons}\end{array}\right]$$

$$-\frac{1}{2}\left[\begin{array}{l}\text{Total number}\\ \text{of bonding}\\ \text{electrons}\end{array}\right]$$

Sol.: Formal charge on S in 

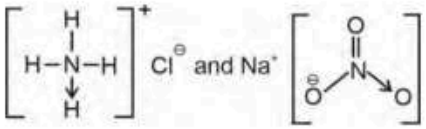
$$= \left[6 - 0 - \frac{1}{2}\times 12\right] = 0$$

Formal charge on central O in 

$$= \left[6 - 2 - \frac{1}{2}\times 6\right] = 1$$

52. Answer (2)

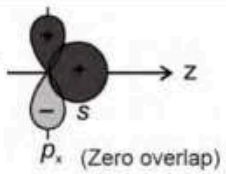
Hint: Coordinate bond is formed by sharing a lone pair to the vacant orbital of another atom.

Sol.:  $Cl^{\ominus}$ and Na^+

Contain both co-ordinate bond and ionic bonds.

53. Answer (2)

Hint: In zero overlap, net overlap of orbital is zero.

Sol.: • 

• (a) and (b) are negative overlap.

• (c) is the presentation of positive overlap.

54. Answer (2)

Hint: For triatomic species if the central atom is *sp* or sp^3d hybridised, the species will be linear in shape.

Sol.:

Species	Hybridisation of central atom	Shape
$[I_3]^+$	sp^3	Bent
SO_2	sp^2	Bent
$BeCl_2$	*sp*	Linear
NO_2^+	*sp*	Linear
H_2O	sp^3	Bent
$[ICl_2]^-$	sp^3d	Linear
CO_2	*sp*	Linear

55. Answer (1)

Hint: For same bond order, large is the size of atom, larger is the bond length.

Sol.: Bond order of NO = $\frac{1}{2}(N_b - N_a)$

$$= \frac{1}{2}(10-5) = 2.5$$

Molecule	Bond energy (kJ mol^{-1})
B_2	293
F_2	155

56. Answer (2)

Hint: It is not necessary that only half filled orbitals participate in hybridisation.

Sol.: Promotion of electron is not an essential condition prior to hybridisation.

57. Answer (2)

Hint: In general ionisation enthalpy increases along the period.

Sol.: • For stable electronic configuration the ionisation enthalpy of the element is higher than the element next to it in a period.

- For stable electronic configuration the ionisation enthalpy of N is higher than O.

58. Answer (4)

Hint: In Doberiener's triads, the middle element of each of the triads has an atomic weight about half way between the atomic weights of the other two.

Sol.: The Dobereiner's Triads are

- Li, Na, K
- Ca, Sr, Ba
- Cl, Br, I

59. Answer (1)

Hint: f-block elements are called inner transition elements.

Sol.: • Bismuth is a metal while tellurium is a semi-metal.

- *s* and *p* block elements are main group elements.
- Vanadium is a *d*-block element.

60. Answer (2)

Hint: Along the period, electronegativity generally increases.

Sol.:

Element	Electronegativity values (on Pauling scale)
Be	1.5
B	2.0
C	2.5
Mg	1.2

61. Answer (4)

Hint: N_2O is a neutral oxide.

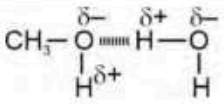
Sol.:

Compound	Chemical Nature
As_2O_3	Amphoteric
SO_2	Acidic
CaO	Basic

62. Answer (2)

Hint: Methyl alcohol (CH_3OH) can form hydrogen bond with each other and also with water.

Sol.: Solubility of CH_3OH in water is explained by intermolecular hydrogen bonding.



63. Answer (3)

Hint: Electronic configuration of B_2 :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^1 = \pi 2p_y^1)$

Sol.: Electronic configuration of B_2^- :

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^1)$

64. Answer (3)

Hint: hybridization of I in IF_7 is sp^3d^3.

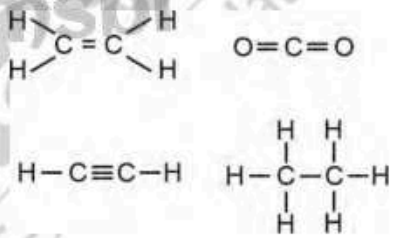
Sol.:

Molecule	Hybridisation of central atom	Shape
IF_5	sp^3d^2	Square pyramidal
IF_7	sp^3d^3	Pentagonal bipyramidal

65. Answer (4)

Hint: If the hybridisation of carbon is sp^3, the hydrocarbon will not form a π-bond.

Sol.:

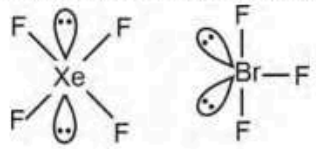


In C_2H_6, only σ – bonds are present.

66. Answer (2)

Hint: Hybridization of Br in BrF_3 is sp^3d.

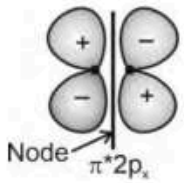
Sol.:



67. Answer (1)

Hint: Molecular orbitals obtained from $2p_x$ and $2p_y$ orbitals are not symmetrical around the bond axis.

Sol.:



π^* antibonding MO has a node between the nuclei.

68. Answer (3)

Hint: If hybridisation of the central atom and shape of the species are same then they are called isostructural species.

Sol.:

Species	Total no of electron	Hybridisation of central atom	Shape
NO_3^-	32	sp^2	Trigonal planar
CO_3^{2-}	32	sp^2	Trigonal planar
XeF_2	72	sp^3d	Linear
IF_2^-	72	sp^3d	Linear
ClO_4^-	50	sp^3	Tetrahedral
PO_4^{3-}	50	sp^3	Tetrahedral
NF_3	34	sp^3	Pyramidal
ClF_3	44	sp^3d	T-shape

69. Answer (1)

Hint: Along the period, atomic size decreases

Sol.: • Down the group, atomic size generally increases.

- Correct order of atomic size

  Si > P > S > C > N

70. Answer (1)

Hint: IUPAC official name of element having atomic number 105 is Dubnium

Sol.:

Atomic Number	IUPAC official name
108	Hassium
102	Nobelium
104	Rutherfordium

71. Answer (1)

Hint: In OF_2 oxidation state of F is –1

Sol.: In OF_2 oxidation state of O will be + 2 whereas in Na_2O, the oxidation state of Na is +1 hence oxidation state of O will be –2.

72. Answer (3)

Hint: According to modern periodic law, the properties of elements are periodic function of their atomic numbers.

Sol.: Neptunium having atomic number 93 belongs to actinoid series.

73. Answer (3)

Hint: Dipole moment depends on bond moments, orbital dipole and shape of the molecules.

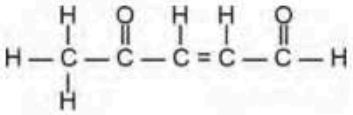
Sol.:

Molecules	Dipole moment (D)
NH_3	1.47
NF_3	0.23
$CHCl_3$	1.04
CCl_4	0

74. Answer (3)

Hint: –CHO contains one π bond

Sol.:



Number of σ bonds = 12

Number of π bonds = 3

75. Answer (4)

Hint: Electronic configuration of N_2 :

$$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^2$$

Sol.: Electronic configuration of N_2^+ :

$$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^1$$

Number of bonding electrons = 9

- Electronic configuration of F_2 :

$$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2$$
$$(\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^2 = \pi^* 2p_y^2)$$

Number of antibonding electrons = 8

76. Answer (2)

Hint: Percentage ionic character = $\frac{\mu_{observed}}{\mu_{calculated}} \times 100$

Sol.: 1D = 3.33×10^{-30} Cm

Percentage ionic character

$$= \frac{1.5 \times 3.33 \times 10^{-30} \times 100}{120 \times 10^{-12} \times 1.6 \times 10^{-19}} = 26\ \%$$

77. Answer (4)

Hint: Smaller the size of atoms, smaller is the bond length.

Sol.:

Bond type	Covalent bond length (pm)
O — H	96

C — H	107
C = N	138
C = O	121

78. Answer (2)

Hint: NO_2 is an odd electron molecule

Sol.: • Octet rule is not satisfied for all the atoms of NO_2 $\ddot{O}=\overset{\oplus}{N}-\ddot{O}:^{\ominus}$

• In SCl_2, Octet rule is obeyed by each atom $:\ddot{Cl}-\ddot{S}-\ddot{Cl}:$

79. Answer (2)

Hint: If net-dipole moment is non-zero then the molecule will be polar

Sol.:

Molecule	Shape	Dipole moment (μ)
PCl_3	Pyramidal (Non-planar)	$\mu \neq 0$
H_2O	Bent (planar)	$\mu \neq 0$
$CHCl_3$	Tetrahedral (Non-planar)	$\mu \neq 0$
NF_3	Pyramidal (Non-planar)	$\mu \neq 0$
BF_3	Trigonal planar	$\mu = 0$

80. Answer (2)

Hint: Bond order = $\frac{1}{2}(N_b - N_a)$

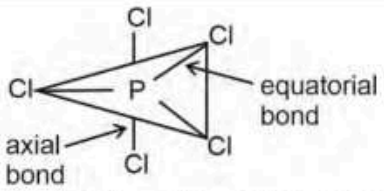
Sol.:

Species	Bond order
O_2	$\frac{1}{2}(10-6)=2$
O_2^-	$\frac{1}{2}(10-7)=1.5$
O_2^{2-}	$\frac{1}{2}(10-8)=1$
O_2^+	$\frac{1}{2}(10-5)=2.5$

81. Answer (4)

Hint: P in PCl_5 in sp^3d hybridised

Sol.:

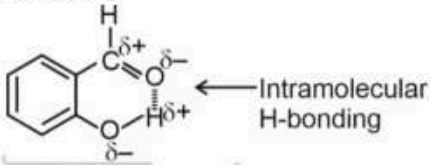


- Axial bond pairs suffer more repulsive interaction from the equatorial bond pairs, therefore axial bonds have been found to be slightly longer and hence slightly weaker than the equational bonds.
- Maximum four atoms lie in a plane

82. Answer (3)

Hint: Intramolecular hydrogen bond is formed when hydrogen atom is in between the two highly electronegative (F, O, N) atoms present within the same molecule.

Sol.:



83. Answer (1)

Hint: If bond order is zero then the species does not exist

bond order = $\frac{1}{2}(N_b - N_a)$.

Sol.: • Bond order of He_2 and Be_2 are zero.

• B_2 : $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2\left(\pi 2p_x^1 = \pi 2p_y^1\right)$. It is paramagnetic in nature due to the presence of 2 unpaired electrons.

84. Answer (4)

Hint: Shape of XeF_4 is square planar

Sol.:

Molecule	Shape	Hybridization of central atom
SF_4	See-saw	sp^3d
XeF_4	Square planar	sp^3d^2
XeO_3	Pyramidal	sp^3
SO_2	Bent	sp^2

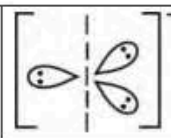
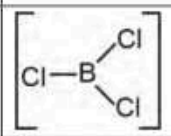
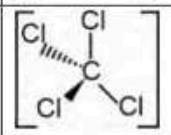
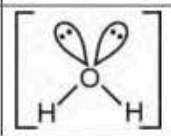
85. Answer (1)

Hint: In I_3^-, central I atom is sp^3d hybridised.

Sol.:

Species	Arrangement of electron pairs	Bond angle

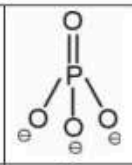
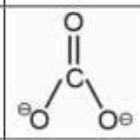
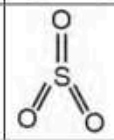
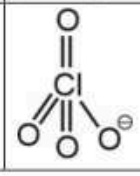
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I_3^-		180°
BCl_3		120°
CCl_4		109.5°
H_2O		104.5°

86. Answer (4)

Hint: The elements which contain d-orbitals can form pπ = dπ bond

Sol.:

•		One pπ – dπ bond between P and O
•		It contains only pπ – pπ bond
•		It contains one pπ – pπ bond and two pπ – dπ bonds.
•		It contains one pπ – dπ bond.

87. Answer (3)

Hint: Electronic configuration of element with atomic number 49 is $[Kr]4d^{10}5s^25p^1$

Sol.: The element is Indium and it belongs to 5th period and 13th group.

88. Answer (3)

Hint: IUPAC name of the element having atomic number 101 is Unnilunium.

Sol.:

Atomic number	Name according to IUPAC nomenclature
101	Unnilunium
109	Unnilennium

89. Answer (4)

Hint: Samarium is lanthanoid metal

Sol.:

Vanadium	–	Transition metal
Samarium	–	Lanthanoid
Chlorine	–	Non-metal
Germanium	–	Metalloid

90. Answer (1)

Hint: For alkali metals, down the group, negative electron gain enthalpy decreases.

Sol.:

Element	$\Delta_{eg}H$ (kJ mol^{-1})
Na	– 53
K	– 48
S	– 200
Se	– 195

[BOTANY]

91. Answer (3)

Hint: The mentioned phase is diplotene

Sol.: Diplotene shows the appearance of chiasmata.

92. Answer (3)

Hint: M phase is the most dramatic period of the cell cycle

Sol.: M phase involves major reorganisation of virtually all components of the cell.

93. Answer (1)

Hint: The life cycle begins with zygote.

Sol.: Any sexually reproducing organism starts its life cycle from a single-celled zygote (diploid).

94. Answer (1)

Hint: Cell division restores the nucleo-cytoplasmic ratio.

Sol.: Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.

95. Answer (3)

Sol.: Biological names are generally in Latin.

96. Answer (3)

Hint: *Solanum nigrum* is the scientific name of makoi.

Sol.: *familiaris* belongs to genus *Canis*.

97. Answer (4)

Hint: It forms at the end of meiosis II.

Sol.: Cytokinesis that follows telophase II results in the formation of a tetrad of cells.

98. Answer (4)

Hint: Meiosis involves only a single cycle of DNA replication.

Sol.: Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S-phase. Cell division does not stop with the formation of the mature organism but continues throughout its life cycle.

99. Answer (4)

Hint: The scientific name of the organism is constituted of specific epithet and generic name.

Sol.: Scientific name gives description about species and genus of the organism along with the author name.

100. Answer (4)

Hint: Only a single cycle of DNA replication is observed during meiosis.

Sol.: There is no replication of DNA during interkinesis.

101. Answer (2)

Hint: Cell plate represents the middle lamella between the walls of two adjacent cells.

Sol.: The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. Chiasmata is the site of crossover. During prophase, the two asters together with spindle fibres forms mitotic apparatus.

102. Answer (2)

Hint: ICZN provides the naming principles for animals.

Sol.: Animals are eukaryotes.

103. Answer (3)

Hint: *Panthera leo* is the name of lion.

Sol.: The correct scientific name of leopard is *Panthera pardus*.

104. Answer (1)

Hint: It occurs in the final stage of meiotic prophase I.

Sol.: By the end of diakinesis, the nucleolus disappears and the nuclear envelope also breaks down.

105. Answer (4)

Hint: The sequence of events by which a cell duplicates its genome, synthesising the other constituents of the cell and eventually divides into two daughter cells is termed as cell cycle.

Sol.: The period of cell cycle varies from organism to organism and from cell to cell of an organism.

106. Answer (3)

Hint: Muscidae belongs to Diptera.

Sol.: The animal order, Carnivora, includes families like Felidae and Canidae.

107. Answer (3)

Hint: Poales is the order to which wheat belongs.

Sol.: Plant families like Convolvulaceae and Solanaceae are included in the order Polymoniales.

108. Answer (3)

Hint: Crossing over is completed by the end of third substage of prophase I.

Sol.: Crossing over takes place in pachytene.

109. Answer (2)

Hint: Telophase is the final stage of karyokinesis.

Sol.: Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

110. Answer (2)

Hint: Both cat and gorilla belong to class mammalia.

Sol.: Both cat and gorilla share three taxonomical categories, namely class (Mammalia), Phylum (Chordata) and Kingdom (Animalia).

111. Answer (2)

Hint: Taxonomy includes classification identification and nomenclature.

Sol.: The taxonomic categories are distinct biological entities and not merely morphological aggregates.

112. Answer (1)

Hint: Centripetal mode of cytokinesis can be seen in animal cells.

Sol.: In animal cell, the cytokinesis is facilitated by furrow formation.

113. Answer (4)

Hint: G_0 phase of the cell cycle is metabolically active.

Sol.: Cells enter the G_1 phase post the G_0 phase.

114. Answer (4)

Hint: In all organisms, the cell cycle is divided into two basic phases.

Sol.: Onion root tip cell has both interphase and M phase. In some lower plants and in some social insects haploid cells also divide by mitosis.

115. Answer (2)

Hint: Lower the taxa, more are the characteristics that the members within the taxon share.

Sol.: Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

116. Answer (1)

Hint: Taxa can indicate categories of very different levels.

Sol.: Dog is a mammal and mammals are animals. Therefore, 'animals', 'mammals', and 'dogs' represent taxa at different levels.

117. Answer (4)

Hint: Carolus Linnaeus proposed binomial nomenclature.

Sol.: Ernst Mayr gave the definition of biological concept of species.

118. Answer (2)

Hint: It marks the beginning of the second phase of mitosis.

Sol.: The complete disintegration of the nuclear envelope marks the start of metaphase.

119. Answer (4)

Hint: Splitting of centromere ensures proper segregation of chromatids to each daughter cell.

Sol.: In animal cells, cytokinesis occurs in a centripetal manner.

120. Answer (3)

Hint: Cell division is a process by which cell divides in two.

Sol.: The sequence of events by which a cell duplicates its genome, synthesise the other constituents of the cell and eventually divided into two daughter cells is termed cell cycle.

121. Answer (2)

Hint: Ernst Mayr has been called 'The Darwin of the 20th century'.

Sol.: Ernst Mayr was awarded with the triple crown of biology.

122. Answer (1)

Hint: Generally, order and other higher taxonomic categories are identified based on the aggregates of characters.

Sol.: Gibbons, based on the common features like presence of notochord and dorsal hollow neural system, are included in phylum chordata.

123. Answer (1)

Hint: A mitotic metaphase chromosome is a typical X-shaped structure.

Sol.: A mitotic metaphase chromosome are made up of two sister chromatids held together by a centromere.

124. Answer (4)

Hint: Meiosis II occurs in haploid cells.

Sol.: On completion of meiosis II, the produced daughter cells have a different genetic composition than the parents.

125. Answer (2)

Hint: The number of pollen grains formed = n/4.

Sol.: The number of pollen grains formed = 100/4 = 25.

126. Answer (2)

Hint: The correct taxonomical hierarchy is as follows:

Species → Genus → Family → Order → Class → Division → Kingdom.

Sol.: Taxonomical categories in ascending order are as follows:

Triticum→	Poaceae→	Poales→	Monocotyledonae→	Angiospermae
(Genus)	(Family)	(Order)	(Class)	(Division)

127. Answer (1)

Hint: Cell cycle in human completes in a day.

Sol.: Human cell in culture can progress through the cell cycle in only about 24 hours.

128. Answer (2)

Hint: DNA replication takes place in S-phase.

Sol.: G_1 phase occurs before S-phase, hence duplicated DNA is not seen here.

129. Answer (1)

Hint: Rudolf Virchow proposed *'Omnis cellula-e-cellula'*.

Sol.: During mitosis, the main function of the spindle fibres is to separate the chromosomes.

130. Answer (1)

Hint: Terminalisation of chiasmata can be observed during the final stage of prophase I.

Sol.: Terminalisation of chiasmata can be observed during diakinesis.

131. Answer (2)

Hint: Number of centromeres found in telophase are equal to the number of chromosomes found in a daughter cell obtained after cytokinesis.

Sol.: Number of chromosomes in a daughter cell

= Number of centromere

= 24

132. Answer (1)

Hint: As we go higher from species to kingdom, the number of common characteristics goes on decreasing.

Sol.: • Diptera (order) > Insecta (Class) (in terms of number of common characters)

• Chordata (Phylum) > Mammalia (Class) (in terms of number of members).

• Muscidae (Family) < Dicotyledonae (Class) (in terms of rank in taxonomical hierarchy)

• Anacardiaceae (Family) > Sapindales (Order) (in terms of number of common characters)

133. Answer (4)

Hint: Mitosis restores the nucleo-cytoplasmic ratio.

Sol.: G_1 phase – Synthesis of proteins, RNA and ATP, except DNA resulting in cell growth.

S phase – Replication of DNA (polymer of deoxyribonucleotides)

G_2 phase – Synthesis of tubulin.

M phase – Helps in restoration of nucleo-cytoplasmic ratio after cell growth.

134. Answer (1)

Hint: A – Family, B – Order, C – Class, D – Division/phylum

Sol.: The similar characters found in the members of an order are less in number as compared to different genera included in a family. Order primata is placed in class mammalia along with order carnivora.

135. Answer (3)

Hint: c-Prophase

e-Metaphase.

Sol.: The correct sequence of events is

$c \rightarrow e \rightarrow a \rightarrow b \rightarrow d$

[ZOOLOGY]

136. Answer (4)

Hint: TV is approximately 500 mL.

Sol.: EC = TV + ERV = 1500 – 1600 mL.

So, value of expiratory capacity is almost equal to three times the tidal volume (TV).

Inspiratory capacity = 3000 – 3500 mL

IRV = 2500 – 3000 mL

RV = 1100 – 1200 mL

137. Answer (3)

Hint: Eliminate the one which is released out.

Sol.: Oxygen (O_2) is utilized by the organisms to indirectly breakdown simple molecules like glucose, amino acids, fatty acids, *etc.*, to derive energy to perform various activities. CO_2 is released during catabolic reactions.

138. Answer (4)

Hint: Partial pressure of gases affects diffusion.

Sol.: Total pressure of a gaseous mixture is the sum of partial pressures of different gaseous constituents present in the gaseous mixture and remains the same for all the gases. So, total pressure of a gaseous mixture is not responsible for their differential diffusion. Partial pressure of different gases is responsible for their different rates of diffusion.

139. Answer (1)

Hint: Division of respiratory tract after bronchi.

Sol.: Only trachea, bronchi, and initial bronchioles of the conducting part of human's respiratory system are supported by incomplete cartilaginous rings. Nasal passage, pharynx, and terminal bronchioles are not supported by incomplete cartilaginous rings.

140. Answer (2)

Hint: Diffusion of gases depends upon their solubility.

Sol.: As the solubility of CO_2 is 20–25 times higher than that of O_2, the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2.

141. Answer (4)

Hint: 1 $mm^{-3} = 10^{-3}$ mL.

Sol.: A healthy individual has 12–16 gms of haemoglobin in every 100 mL (1 decilitre) of blood. 5.5 million RBCs are present in 0.001 mL of blood and 55 million RBCs are present in 0.01 mL of blood.

142. Answer (3)

Hint: These cells are devoid of nucleus.

Sol.: In most of the mammals including humans, RBCs are devoid of nucleus and are maximum in number among other formed elements. Neutrophils are maximum in number among WBCs and granulocytes, whereas lymphocytes are maximum in number among agranulocytes.

143. Answer (4)

Hint: Rh factor of foetal RBCs is immunogen for mother.

Sol.: Erythroblastosis foetalis, a condition in which mother starts to produce anti-Rh antibodies against Rh factor of foetal blood as mother is Rh-ve, so, anti-Rh antibodies are administered to mother to destroy Rh antigen immediately to prevent mother's immune system to activate against Rh-antigen.

144. Answer (3)

Hint: Lymph is also a fluid connective tissue circulating in lymphatic system.

Sol.: Some water along with water soluble substances which filter out in the region of capillary bed is called tissue fluid. It is devoid of most of the formed elements, and few larger proteins present in blood. It is responsible for transportation of nutrients and is also a carrier for hormones. It contains lymphocytes but lacks RBCs.

145. Answer (1)

Hint: RBCs undergo clumping due to transfusion of mismatched blood.

Sol.: Thrombokinase is produced by cascade processes which is responsible for conversion of prothrombin into thrombin. Clot is mainly formed by a network of threads called fibrins in which formed elements of blood are trapped.

146. Answer (3)

Hint: Additional muscles in the abdomen help in forceful breathing.

Sol.: Normal breathing is performed by contraction and relaxation of diaphragm and external inter-costal muscles. Contraction of internal inter-costal muscles and abdominal muscles help in forceful expiration. Muscles in the abdomen have ability to increase the strength of forceful inspiration as well as expiration.

147. Answer (4)

Hint: Factors are favourable for diffusion of CO_2 in opposite direction than O_2.

Sol.: Rate of diffusion of gases depends upon partial pressure of gases, thickness of diffusion membrane, and solubility of gases. All these factors facilitate diffusion of O_2 from alveoli to deoxygenated blood and from oxygenated blood to tissues, whereas diffusion of CO_2 is facilitated from tissues to deoxygenated blood and from deoxygenated blood to alveoli.

148. Answer (2)

Hint: Exclude the function of diffusion membrane.

Sol.: Respiratory system of human beings is differentiated into two parts: conducting part and exchange part. The parts starting with the external nostrils up to the terminal bronchioles constitute the conducting part, whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system. All mentioned functions are associated with conducting part, whereas exchange part provides surface for gaseous exchange.

149. Answer (2)

Hint: 100 mL of oxygenated blood delivers 5 mL of O_2 to tissues under normal physiological conditions.

Sol.: Blood pumped out by left ventricle in each cardiac cycle = Stroke volume = 70 mL

Blood pumped out by left ventricle per minute = Cardiac Output = HR × SV = 72 × 70 $\simeq$ 5000 mL
Because 100 mL of oxygenated blood delivers only 5 mL of O_2 to tissue, so oxygen delivered to tissue per minute = $\frac{5}{100} \times 5000 = 250$ mL.

150. Answer (1)

Hint: Overall decrease in intra-pulmonary pressure causes inspiration.

Sol.: Contraction of diaphragm increases the volume of thoracic chamber in antero-posterior axis and contraction of external inter-costal muscles increases volume of thoracic chamber in dorso-

ventral axis that leads to increase in volume of lungs. As a result, intra-pulmonary pressure decreases below atmospheric pressure to create a suction pull to rush air into lung alveoli from atmosphere *i.e.*, inspiration.

151. Answer (2)

Hint: Animals with nephridia for excretion.

Sol.: The closed circulatory system with a heart to pump blood in blood vessels appeared for the first time in members of the phylum Annelida, and open circulatory system appeared for the first time in members of the phylum Arthropoda.

152. Answer (4)

Hint: Think about pulmonary artery.

Sol.: Arteries run away from the heart and usually carry oxygenated blood with few exceptions, *e.g.*, pulmonary arteries which carry deoxygenated blood. On the other hand, veins run towards the heart and usually carry deoxygenated blood with few exceptions, *e.g.*, pulmonary veins which carry oxygenated blood from lungs to left auricle.

153. Answer (4)

Hint: Fishes exhibit single circulation.

Sol.: Animal 'A' is a fish and 'B' is an amphibian. In fishes, deoxygenated blood received by atrium is pumped to gills for oxygenation and blood passes only once through heart, called single circulation. In amphibians, deoxygenated blood is received by right atrium and oxygenated blood is received by left atrium. Blood from both atria reaches a single ventricle and this mixed blood is pumped to systems of the body; such type of circulation is called incomplete double circulation.

154. Answer (4)

Hint: Ratio of diastolic to systolic blood pressure is 80 : 120 = 2 : 3.

Sol.: pO_2 at which haemoglobin is 75% saturated with O_2 = 40 mm Hg

pO_2 at which haemoglobin is 90% saturated with O_2 is about 60 mm Hg

Ratio of these two values of pO_2 = 40 : 60 = 2 : 3.

Per cent of lymphocytes in the total WBCs = 20 – 25%.

Per cent of CO_2 transported in the form of carbamino-haemoglobin = 20 – 25%.

Transportation of CO_2 dissolved in plasma = 7%.
Transport of CO_2 as bicarbonate in plasma = 70%.

So, per cent of CO_2 transported in plasma as bicarbonate is 10 times to the CO_2 transported in dissolved state in plasma.

155. Answer (1)

Hint: Semilunar valves close when pressure within ventricles decreases.

Sol.: During joint diastole, semilunar valves close and AV valves open and blood freely moves from vena cava to right atrium and from atria to ventricles. Closure of AV valves occurs during ventricular systole. Only AV valves remain open during joint diastole.

156. Answer (1)

Hint: Cardiac output = Heart rate × Stroke volume.

Sol.: Stroke volume = End diastole volume – End systole volume

= [140 – 60] = 80 mL

Cardiac output = Heart rate × Stroke volume

$= \frac{50}{\text{min}} \times 80 \text{ mL} = 4000 \text{ mL/min} = 4 \text{ L/min}$

157. Answer (2)

Hint: Capillaries are connecting link between arterial system and venous system.

Sol.:

Coronary system	–	Associated with blood supply to heart
Hepatic portal vein	–	Drains blood from intestine to liver
Capillary	–	A link between arteriole and venule
Vena cava	–	Largest vein that opens into right atrium of heart

158. Answer (1)

Hint: T-P gap of ECG represents joint diastole.

Sol.: TP gap of ECG represents joint diastole during which heart muscles remain electrically silent. Eosinophils release histaminase which degrades histamine released from basophils and mast cells.

159. Answer (3)

Hint: Elasticity is the property shown by elastic fibres and muscle fibres.

Sol.: Tunica media is a middle layer of blood vessels with smooth muscle and elastic fibres. Both

muscle fibres and elastic fibres of connective tissue exhibit a property called elasticity.

160. Answer (2)

Hint: Constitute minimum per cent of the total WBCs.

Sol.: Basophils and mast cells release histamine which participate in inflammatory reactions of the body. Cortisol is responsible for anti-inflammatory reactions of the body.

161. Answer (4)

Hint: It is situated close to atrio-ventricular septum.

Sol.: SAN (Sino-Atrial Node) is a patch of nodal tissue present in the right upper corner of right atrium and AVN (Atrio-Ventricular Node) is another patch of nodal tissue situated in lower left corner of the right atrium.

162. Answer (4)

Hint: Function of sympathetic and parasympathetic neural system is antagonistic to each other.

Sol.: A specialised neural centre in the medulla oblongata can moderate the cardiac functions through autonomic nervous system. Neural signals through the sympathetic nerve fibres can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output. On the other hand, parasympathetic neural signals decrease the rate of heart beat, speed of conduction of action potentials, the strength of ventricular systole and thereby the cardiac output. Adrenal medullary hormones also perform the function of sympathetic neural system.

163. Answer (3)

Hint: The unique shape of mammalian RBCs relates to their function of transporting O_2.

Sol.: Mammalian RBCs are superior than RBCs of other vertebrates because:

(i) Biconcave shape of mammalian RBCs provides more surface area for diffusion of gases.

(ii) Absence of nucleus, mitochondria, *etc.*, provide large volume for storage of respiratory chromoprotein called haemoglobin.

164. Answer (4)

Hint: Universal recipient individuals lack antibodies against all blood group antigens.

Sol.: AB+ve blood group individuals are considered as universal recipient because antibodies against blood group antigens are not present in their blood plasma. Serum contains antibodies but lacks clotting factors. Blood pressure during ventricular systole is maximum in aorta due to its minimum total cross-section area and minimum in capillaries due to their maximum total cross-section area.

165. Answer (3)

Hint: Exclude the pacemaker of heart.

Sol.: SA node generates impulses at maximum rate and is considered as pacemaker of heart. AV node has minimum conduction velocity so that it synchronises atrial and ventricular systole in a synchronised order. So, AV node is considered as pace setter of the heart. AV Bundle transmits impulses from AVN of atria to the ventricles and electrically connects atria to ventricles for transmission of impulses.

166. Answer (4)

Hint: Intra-pleural pressure decreases on increasing volume of thoracic cavity.

Sol.: Normal inspiration takes place due to contraction of diaphragm that leads to increase in diameter/length of thoracic cavity in antero-posterior axis. Due to downward movement of diaphragm, the volume of thoracic cavity and thus intra-pulmonary volume increases, and that of abdominal cavity decreases, increasing intra-abdominal pressure.

167. Answer (2)

Hint: Chemosensitive area in peripheral and central chemoreceptors are sensitive to H^+ and CO_2.

Sol.: Change in concentration of H^+ and CO_2 in arterial oxygenated blood affects chemoreceptors of aortic arch and carotid body. Similarly, change in concentration of H^+ and CO_2 in CSF affects the chemosensitive area present in medulla oblongata. Variation in arterial Na^+ concentration does not affect respiration.

168. Answer (2)

Hint: Decrease in CO_2 shifts oxygen dissociation curve to left side.

Sol.: The factors which favour oxygenation of haemoglobin to increase its percentage saturation are:

(i) Decrease in pCO_2

(ii) Decrease in temperature

(iii) Increase in pH

(iv) Decrease in H^+ concentration

(v) Decrease in 2,3-BPG

169. Answer (3)

Hint: The dome-shaped structure forms floor of thoracic cavity.

Sol.: The chamber formed dorsally by the vertebral column, ventrally by sternum, laterally by ribs, and on the lower side by dome-shaped diaphragm is called thoracic cavity.

170. Answer (2)

Hint: Atherosclerosis leads to coronary artery disease.

Sol.: Asthma is an allergic disease characterised by wheezing during respiration due to inflammation of bronchi and bronchioles. In emphysema, respiratory surface decreases due to damage of alveolar walls. Lung fibrosis occurs in occupational lung diseases such as asbestosis. Coronary artery disease is associated with taking fat rich diet in which deposition of fat, cholesterol, calcium in lining of blood vessels leads to narrowing of their lumen.

171. Answer (1)

Hint: Stimulation of chemoreceptors in carotid bodies increases breathing rate.

Sol.: Stimulation of chemoreceptors in carotid bodies increases breathing rate. Injection of drug that activates chemoreceptors in carotid bodies would be expected to increase breathing rate and thus increase in the pO_2 of arterial blood. In CO is poisoning, CO tightly bound to haemoglobin so that dissociation of oxyhaemoglobin as well as binding of O_2 with haemoglobin becomes poor.

172. Answer (3)

Hint: Animals with tissue level of organisation.

Sol.: Simple organisms like sponges and coelenterates circulate water from their surroundings through their body cavities to facilitate the cells to exchange the substances.

173. Answer (2)

Hint: Cells which constitute maximum per cent of the total WBCs.

Sol.: Neutrophils are polymorphonuclear leucocytes which constitute 60-65% of the total WBCs. Maximum per cent of blood corpuscles is formed by RBCs.

174. Answer (3)

Hint: Number of QRS complexes in a given time represents heart beat rate.

Sol.: To obtain a standard ECG, a patient is connected to machine with three electrical leads (one to each wrist and to the left ankle). Letters of ECG from P to T corresponds to a specific electrical activity of heart. By counting the number of QRS complexes that occur in a given time period, one can determine heart beat rate of an individual.

175. Answer (3)

Hint: VC = TLC – RV.

Sol.: Vital capacity = TV + IRV + ERV

= 500 + (2500-3000) mL + (1000-1100) mL

= 4100 – 4600 mL.

176. Answer (2)

Hint: Intra-pleural pressure is less than atmospheric pressure

Sol.: After forceful expiration, residual volume left in lungs, cannot be expelled out due to negative intra-pleural pressure which pulls lung walls to outside. So, it prevents elastic recoiling of lungs and thus prevents alveoli to collapse.

177. Answer (2)

Hint: Haemoglobin is saturated upto 50% when pO_2 is less than 40 mmHg.

Sol.: P_{50} value of haemoglobin is about 28 mmHg in the range of 25-30 mmHg. Increase in P_{50} value decreases affinity of Hb for O_2 and *vice versa*.

178. Answer (2)

Hint: Left side of heart pumps blood to systems of the body.

Sol.: Right atrium and left ventricle are included in systemic circulation as former receives deoxygenated blood from systems of the body and left ventricle pumps oxygenated blood to systems of the body. Similarly, right ventricle and left atrium are parts of pulmonary circulation because right ventricle pumps deoxygenated blood to lungs and left atrium receives oxygenated blood from lungs.

179. Answer (2)

Hint: In cardiac arrest, heart completely stops its function.

Sol.:

Angina pectoris	–	Acute chest pain due to low supply of O_2 to heart muscles.

Heart failure	–	Heart is not pumping blood effectively as needs of the body.
Cardiac arrest	–	Heart stops beating.
Heart attack	–	Heart muscles are suddenly damaged due to low O_2 supply.
Hypertension		BP > 140/90 of a patient on repeated checks.

180. Answer (4)

Hint: 'Dub' sound is produced due to closure of semilunar valves.

Sol.: 100% filling of ventricles occurs at the end of ventricular diastole after atrial systole. Heart sound 'Lub' is produced at the beginning of ventricular systole due to closure of AV valves whereas 'dub' sound is produced at the beginning of ventricular diastole due to closure of semilunar valves.

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